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Prostate-specific antigen (PSA) test for prostate cancer screening (Review)

Franco JVA, Hwang EC, Jung JH, Vaimberg O, Ilic D, Cleves A, Dahm P

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[Intervention Review]

Prostate-specific antigen (PSA) test for prostate cancer screening

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ABSTRACT

Rationale

Prostate cancer screening remains controversial. A 2013 update of this Cochrane review concluded that prostate cancer screening does not significantly reduce disease-specific or overall mortality. Since then, results from two new, large trials have become available, alongside extended follow-up data from previously included trials. Thus, an update was warranted.

Objectives

To assess the effects of prostate cancer screening compared to no screening in men with no prior diagnosis of prostate cancer.

Search methods

We searched the Cochrane Library, MEDLINE, Embase, five other databases, and two trial registers from their inception to November 2025. We did not apply language restrictions.

Eligibility criteria

We included randomised controlled trials (RCTs) of screening versus no screening for prostate cancer.

Outcomes

Critical outcomes were prostate cancer-specific mortality, overall mortality, and adverse events. Important outcomes included quality of life, prostate cancer diagnosis, localised (stage I and stage II) prostate cancer diagnosis, advanced (stage III and IV) prostate cancer diagnosis, and metastatic prostate cancer diagnosis.

Risk of bias

We used the Cochrane risk of bias tool to assess the risk of bias in the studies.

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Synthesis methods

Two review authors independently selected trials, extracted data, and assessed risk of bias. We used a random-effects meta-analysis model to combine data. We report dichotomous data as incidence rate ratios when outcomes were reported as person-years; otherwise, we report risk ratios (RRs) with 95% confidence intervals (CIs). We assessed the certainty of the evidence using GRADE and applied thresholds for minimal clinically important differences (MCIDs).

Included studies

We included six RCTs with a total of 789,086 randomised participants. The trials were conducted in Europe and North America, and involved men aged 45 to 80 years. Five trials compared prostate-specific antigen (PSA) testing, with or without digital rectal examination (DRE), to usual care without systematic PSA screening. The three largest studies, accounting for 85% of randomised participants, were the CAP trial (UK), PLCO trial (USA), and ERSPC (Europe). One trial with preliminary results compared PSA-based screening in combination with kallikrein and magnetic resonance imaging (MRI) with no screening. Follow-up ranged from 3.2 to 23 years.

Synthesis of results

Screening based on PSA (alone) versus no screening

These results are based on the certainty of evidence for each outcome and are therefore mainly derived from a sensitivity analysis by risk of bias, using the ERSPC trial.

Screening likely results in a reduction in prostate cancer-specific mortality (sensitivity analysis from ERSPC: Rate Ratio 0.87, 95% CI 0.80 to 0.95; 1 study, 162,236 participants; moderate-certainty evidence; MCID: 1 per 1000 participants). Assuming a baseline risk of 16 prostate-cancer-related deaths per 1000, this corresponds to 2 fewer (95% CI 3 fewer to 1 fewer) per 1000.

Based on the analysis that included all eligible trials, screening may reduce overall mortality, although the confidence interval includes the possibility of little to no effect (Rate Ratio 0.99, 95% CI 0.97 to 1.00; I^2 = 22%; 4 studies, 675,121 participants; low-certainty evidence; MCID: 1 per 1000 participants). Assuming a baseline risk of 491 deaths from any cause per 1000, this corresponds to 5 fewer (95% CI 15 fewer to 0 fewer) per 1000.

Screening may result in little to no difference in adverse events (RR 1.32, 95% CI 0.48 to 3.65; 1 study, 408,721 participants; low-certainty evidence; MCID: 1 per 1000 participants), as assessed by deaths related to the prostate biopsy or prostate cancer treatment. Assuming a baseline risk of 3 deaths from any cause per 100,000, this corresponds to 1 more (95% CI 2 fewer to 8 more) per 100,000.

Screening may result in little to no difference in quality of life (SF-6D range 0 to 1; higher values indicate higher quality of life; MCID 0.1; screening 0.777 versus control 0.779; P = 0.88; 1 study, 969 participants; low-certainty evidence).

Screening likely increases prostate cancer diagnoses (sensitivity analysis data: Rate Ratio 1.30, 95% CI 1.27 to 1.34; 1 study, 162,241 participants; moderate-certainty evidence; MCID of 10 per 1000 participants), including localised (stage I and II) disease (sensitivity analysis data: RR 1.53, 95% CI 1.48 to 1.59; 1 study, 162,236 participants; moderate-certainty evidence; MCID: 10 per 1000 participants). Screening likely has little to no effect on advanced (stage III and IV) prostate cancer diagnosis (sensitivity analysis data: RR 0.90, 95% CI 0.85 to 0.95; 1 study, 162,236 participants; moderate-certainty evidence; MCID: 10 per 1000 participants). It may reduce metastatic prostate cancer diagnosis (sensitivity analysis data: RR 0.65, 95% CI 0.59 to 0.71; 1 study, 162,236 participants; low-certainty evidence; MCID: 5 per 1000 participants).

Screening based on PSA, kallikrein panel, and MRI versus control

Prostate cancer-specific mortality, overall mortality, adverse events, quality of life, and metastatic prostate cancer diagnosis were not reported in early and preliminary results of the included study. Screening based on PSA, kallikrein panel, and MRI likely has little to no effect on prostate cancer diagnosis compared to control (RR 1.85, 95% CI 1.56 to 2.19; 1 study, 60,745 participants; moderate-certainty evidence; MCID: 10 per 1000 participants). It likely also has little to no effect on localised prostate cancer diagnoses (RR 1.88, 95% CI 1.55 to 2.27; 1 study, 60,745 participants; moderate-certainty evidence; MCID: 10 per 1000 participants) and advanced prostate cancer diagnoses (RR 1.78, 95% CI 1.19 to 2.66; 1 study, 60,745 participants; moderate-certainty evidence; MCID: 10 per 1000 participants).

Authors' conclusions

Screening likely reduces prostate cancer-specific mortality and may reduce overall mortality. It may have little to no effect on adverse events (as measured by intervention-related mortality). Study limitations, inconsistency, and imprecision reduced our confidence in the effect estimates. Interpretation of these findings is highly sensitive to the choice of MCID. We found insufficient evidence on the potential harms of screening, such as biopsy- and treatment-related complications. Emerging alternatives, such as screening with a kallikrein panel and MRI, may have little to no effect on diagnoses of prostate cancer, but the results on mortality are not yet known.

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Registration

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Review update (2013) DOI: 10.1002/14651858.CD004720.pub3

PLAIN LANGUAGE SUMMARY

Is a blood test (PSA) an effective screening method for prostate cancer?

Key messages

- Screening for prostate cancer using a blood test likely reduces the risk of dying from prostate cancer and may also reduce the risk of dying from any cause. Screening likely makes little to no difference in the diagnosis of advanced prostate cancer but may reduce the diagnosis of metastatic prostate cancer (cancer that has spread to other body parts).
- Screening likely increases the number of men diagnosed with any prostate cancer, as well as early-stage cancer, placing some at risk of being diagnosed and treated for cancers that may not matter during their lifetime.
- A newer screening approach that combines the standard blood test with a wider set of blood test markers and prostate imaging likely finds more prostate cancers, but whether it reduces deaths from prostate cancer or any cause is unclear.

What is prostate cancer screening?

Prostate cancer develops in the prostate, a small gland below the bladder that makes fluid for semen. It is one of the most common types of cancer in men, mainly affecting men aged over 50. If caught early, it can usually be treated successfully. Screening tests are designed to diagnose prostate cancer in the early stages where there may be no symptoms or before slow-growing cancers cause problems.

The 'PSA' (prostate-specific antigen) blood test looks for high PSA levels in the blood, which can indicate prostate cancer. The next step for men with high PSA is a biopsy, where a small tissue sample is removed and tested for cancer. Treatments such as surgery or radiation are then given. But these treatments may cause unwanted effects, such as leaking urine or having to urinate too often, difficulties with erections, and bowel problems.

What did we want to find out?

We aimed to find out if screening for prostate cancer using PSA in men without any cancer symptoms reduces their risk of dying from prostate cancer or other causes. We also wanted to find out if PSA testing:

- causes any harm;
- affects men's quality of life; and
- has any impact on how often prostate cancer is diagnosed, and whether diagnoses occur when cancer is localised (only in the prostate gland), advanced (grown beyond the prostate gland), or has spread to other parts of the body (metastasised).

What did we do?

We searched for studies that compared PSA screening to no screening in men with no cancer symptoms. We compared and summarised the results of the studies and rated our confidence in the evidence, based on factors such as study methods and numbers of men included.

What did we find?

We found six studies involving 789,086 men, most of whom were White. The studies were conducted in Canada, Europe, and the USA. The men were between 45 and 80 years old and were followed for 3.2 to 23 years.

Main results

PSA testing:

- likely reduces the risk of dying from prostate cancer over 23 years of follow-up: 2 fewer men die of prostate cancer per 1000 men screened (evidence from 1 study with 162,241 men);
- may reduce the risk of dying from any cause: 5 fewer men die of prostate cancer per 1000 screened (4 studies, 675,121 men);
- may make little to no difference to the number of men who experience serious harmful, unwanted events, defined as deaths from biopsy or prostate cancer treatment (1 study, 408,721 men);
- may result in little to no difference in men's quality of life (1 study, 969 men).

For prostate cancer diagnoses, evidence from 1 trial with 162,236 men shows that PSA testing:

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- likely increases the numbers of men diagnosed with any prostate and localised prostate cancer: 36 more men are diagnosed with any prostate cancer and 34 more men with localised cancer per 1000 men screened;
- likely makes little to no difference to advanced prostate cancer diagnoses;
- may reduce the number diagnosed with metastatic cancer: 5 fewer men per 1000 screened.

We did not find evidence on other unwanted effects of screening, including problems related to biopsies (e.g. serious infections or hospital admissions) or treatments (e.g. urinary problems, erection difficulties, bowel problems).

One study, involving 60,745 men, examined a combination of the standard blood test plus a wider set of blood tests and prostate imaging. This approach likely has little to no effect on the number of men diagnosed with prostate cancer, including both early and advanced stages. This study did not report whether testing affected the number of deaths.

What are the limitations of the evidence?

We are moderately confident in the evidence from one large, well-conducted, long-running study. We have lower confidence in the evidence from the other five studies due to concerns about how they were conducted and differences in their results.

This review provides little information about the unwanted effects of screening because the studies we considered do not typically collect this information, especially delayed or infrequent unwanted effects.

How up to date is this evidence?

This updates our previous review published in 2013. The evidence is current to 19 November 2025.

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Summary of findings 1. Summary of findings table - Prostate-specific antigen (PSA)-based screening compared to no screening for men at risk of prostate cancer

Prostate-specific antigen (PSA)-based screening compared to no screening for men at risk of prostate cancer

Patient or population: men at risk of prostate cancer
Setting: outpatient: studies in Europe (ERSPC, Stockholm), USA (PLCO), UK (CAP), and Canada (Quebec)
Intervention: prostate-specific antigen (PSA)-based screening
Comparison: no screening

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	№ of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with no screening	Risk with prostate-specific antigen (PSA)-based screening				
Prostate cancer-specific mortality (minimal clinically important difference (MCID) 1 per 1000) follow-up: range 11 years to 23 years	Baseline risk from ERSPC trial		**RR 0.91** (0.85 to 0.99)	721607 (5 RCTs)	⊕⊕⊖⊖ Low[b,c]	Screening may reduce prostate cancer-specific mortality.
	16 per 1000[a]	**15 per 1000** (14 to 16)				
Overall mortality (MCID 1 per 1000) follow-up: range 15 years to 23 years	Baseline risk from ERSPC trial		**RR 0.99** (0.97 to 1.00)	675121 (4 RCTs)	⊕⊕⊖⊖ Low[b,c]	Screening may reduce overall mortality although the confidence interval includes the possibility of little to no effect.
	491 per 1000[a]	**486 per 1000** (476 to 491)				
Adverse events (mortality due to diagnostic biopsy or prostate cancer treatment) (MCID 1 per 1000) assessed with: mortality due to diagnostic biopsy or prostate cancer treatment	Baseline risk from CAP trial		**RR 1.32** (0.48 to 3.65)	408721 (1 RCT)	⊕⊕⊖⊖ Low[e,f]	Screening may result in little to no difference in adverse events.
	3 per 100,000[d]	**4 per 100,000** (1 to 11)				
Quality of life (MCID 0.09 to 0.10) assessed with: SF-6D (range 0 to 1, higher values indicate higher quality of life) follow-up: 5 years	SF-6D: screening 0.777 vs control 0.779, p = 0.88			969 (1 RCT)	⊕⊕⊖⊖ Low[g]	Screening may result in little to no difference in quality of life.

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Prostate cancer diagnosis (MCID 10 per 1000) assessed with: biopsy follow-up: range 5 years to 23 years	Baseline risk from ERSPC trial 120 per 1000[a]	**136 per 1000** (122 to 150)	**RR 1.13** (1.02 to 1.25)	675121 (4 RCTs)	⊕⊕⊖⊖ Low[b,h,i]	Screening may increase prostate cancer diagnoses.
Localised (stage I and II) prostate cancer diagnosis (MCID 10 per 1000) assessed with: biopsy, computed tomography, or magnetic resonance imaging follow-up: range 15 years to 23 years	Baseline risk from ERSPC trial 64 per 1000[a]	**81 per 1000** (54 to 123)	**RR 1.27** (0.84 to 1.92)	647640 (3 RCTs)	⊕⊖⊖⊖ Very low[b,h,j]	Screening may increase localised (stage I and II) prostate cancer diagnoses, although the confidence interval includes the possibility of little to no effect, and the evidence is very uncertain.
Advanced (stage III and IV) prostate cancer diagnosis (MCID 10 per 1000) assessed with: biopsy, computed tomography, or magnetic resonance imaging follow-up: range 15 years to 23 years	Baseline risk from ERSPC trial 31 per 1000[a]	**28 per 1000** (27 to 28)	**RR 0.89** (0.88 to 0.91)	647640 (3 RCTs)	⊕⊕⊕⊖ Moderate[b]	Screening likely results in little to no difference in advanced (stage III and IV) prostate cancer diagnoses.
Metastatic prostate cancer diagnosis (MCID 5 per 1000) assessed with: biopsy, computed tomography, or magnetic resonance imaging follow-up: range 16 years to 23 years	Baseline risk from ERSPC trial 13 per 1000[a]	**9 per 1000** (7 to 12)	**RR 0.73** (0.56 to 0.94)	238919 (2 RCTs)	⊕⊕⊖⊖ Low[b,h,i]	Screening may result in little to no difference in metastatic prostate cancer diagnoses.

*__The risk in the intervention group__ (and its 95% confidence interval) is based on the assumed risk in the comparison group and the __relative effect__ of the intervention (and its 95% CI).

CI: confidence interval; **RR:** risk ratio

GRADE Working Group grades of evidence
High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.
Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_455170215978749370.

[a] The control rate was extracted from the 23-year follow-up of the ERSPC trial.
[b] Downgraded one level due to high risk of bias: all studies had an overall high risk of bias.
[c] Downgraded one level due to imprecision: the confidence interval crosses the minimally important difference (1 per 1000).
[d] The control rate was extracted from the 15-year follow-up of the CAP trial.

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[e] Downgraded one level due to high risk of bias: the included study was at an overall high risk of bias.
[f] Downgraded one level due to indirectness: these data represent only part of the spectrum of possible adverse events (including non-fatal sepsis, among others).
[g] Downgraded two levels due to study limitations: lack of allocation concealment, attrition bias, and the exclusion of men with a diagnosis of prostate cancer from this analysis.
[h] Downgraded one level for inconsistency (high I2 value reflects clinically important heterogeneity).
[i] We did not downgrade further for imprecision, as we assumed the apparent imprecision resulted from inconsistency, for which we downgraded the evidence.
[j] Downgraded one level due to imprecision: the confidence interval crosses the minimally important difference (10 per 1000).

Summary of findings 2. Summary of findings table - Prostate-specific antigen (PSA)-based screening compared to no screening for men at risk of prostate cancer (sensitivity analysis)

Prostate-specific antigen (PSA)-based screening compared to no screening for men at risk of prostate cancer (sensitivity analysis)

Patient or population: men at risk of prostate cancer (sensitivity analysis)
Setting: outpatient - Europe (ERSPC trial)
Intervention: prostate-specific antigen (PSA)-based screening
Comparison: no screening

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	№ of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with no screening	Risk with prostate-specific antigen (PSA)-based screening				
Prostate cancer-specific mortality (minimal clinically important difference (MCID) 1 per 1000) follow-up: 23 years[a]	ERSPC trial 16 per 1000[c]	**14 per 1000** (13 to 15)	**RR 0.87** (0.80 to 0.95)	162236 (1 RCT)	⊕⊕⊕⊖ Moderate[b]	Screening likely reduces prostate cancer-specific mortality.
Overall mortality (MCID 1 per 1000) follow-up: 23 years[a]	ERSPC trial 491 per 1000[c]	**491 per 1000** (481 to 501)	**RR 1.00** (0.98 to 1.02)	162236 (1 RCT)	⊕⊖⊖⊖ Very low[b,d]	We are uncertain about the effects on overall mortality.
Adverse events - not reported	-	-	-	-	-	
Quality of life (MCID 0.09 to 0.10) assessed with: SF-6D (range 0 to 1, higher values indicate higher quality of life) follow-up: 5 years	SF-6D: screening 0.777 vs control 0.779, p = 0.80			969 (1 RCT)	⊕⊕⊖⊖ Low[e]	Screening may result in little to no difference in quality of life.

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Prostate cancer diagnosis (MCID 10 per 1000) assessed with: biopsy follow-up: 23 years[f]	ERSPC trial		**RR 1.30** (1.27 to 1.34)	162241 (1 RCT)	⊕⊕⊕⊖ Moderate[b]	Screening likely results in an increase in prostate cancer diagnoses.
	120 per 1000[c]	**156 per 1000** (152 to 161)				
Localised (stage I and II) prostate cancer diagnosis (MCID 10 per 1000) assessed with: biopsy, computed tomography, or magnetic resonance imaging follow-up: 23 years[f]	ERSPC trial		**RR 1.53** (1.48 to 1.59)	162236 (1 RCT)	⊕⊕⊕⊖ Moderate[b]	Screening likely results in an increase in localised (stage I and II) prostate cancer diagnoses.
	64 per 1000[c]	**98 per 1000** (95 to 102)				
Advanced (stage III and IV) prostate cancer diagnosis (MCID 10 per 1000) assessed with: biopsy, computed tomography, or magnetic resonance imaging follow-up: 23 years[f]	ERSPC trial		**RR 0.90** (0.85 to 0.95)	162236 (1 RCT)	⊕⊕⊕⊖ Moderate[b]	Screening likely results in a reduction in advanced (stage III and IV) prostate cancer diagnoses.
	31 per 1000[c]	**28 per 1000** (26 to 29)				
Metastatic prostate cancer diagnosis (MCID 5 per 1000) assessed with: biopsy, computed tomography, or magnetic resonance imaging follow-up: 16.9 years	ERSPC trial		**RR 0.65** (0.59 to 0.71)	162236 (1 RCT)	⊕⊕⊖⊖ Low[b,g]	Screening may reduce metastatic prostate cancer diagnoses.
	13 per 1000	**8 per 1000** (8 to 9)				

*__The risk in the intervention group__ (and its 95% confidence interval) is based on the assumed risk in the comparison group and the __relative effect__ of the intervention (and its 95% CI).

CI: confidence interval; **RR:** risk ratio

GRADE Working Group grades of evidence
High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.
Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_457215620611971863.

[a] Minimal clinically important difference (MCID): absolute change of 0.1% or 1 per 1000 participants.
[b] Downgraded one level due to high risk of bias: although the ERSPC study is at lower risk of bias, which was considered when conducting this sensitivity analysis, it has an overall high risk of bias.
[c] The control rate was extracted from the 23-year follow-up of the ERSPC trial.
[d] Downgraded two levels due to serious imprecision: the confidence interval crosses the minimally important difference (1 per 1000), indicating possible benefits and harms.
[e] Downgraded two levels due to study limitations: lack of allocation concealment, attrition bias, and the exclusion of men with a diagnosis of prostate cancer from this analysis.
[f] Minimal clinically important cifference (MCID): absolute change of 1% or 10 per 1000 participants.

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g Downgraded one level for imprecision: the confidence interval crosses the minimally important difference (5 per 1000), indicating possibly little to no effect.

Summary of findings 3. Summary of findings table - Screening (with PSA, kallikrein, and magnetic resonance imaging) compared to no screening for men at risk of prostate cancer

Screening (with PSA, kallikrein, and magnetic resonance imaging) compared to no screening for men at risk of prostate cancer

Patient or population: men at risk of prostate cancer
Setting: outpatient - Finland (ProScreen study)
Intervention: screening (with PSA, kallikrein, and magnetic resonance imaging)
Comparison: no screening

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	№ of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with no screening	Risk with screening (with PSA, kallikrein, and magnetic resonance imaging)				
Prostate cancer-specific mortality - not reported	-	-	-	-	-	
Overall mortality - not reported	-	-	-	-	-	
Adverse events - not reported	-	-	-	-	-	
Quality of life - not reported	-	-	-	-	-	
Prostate cancer diagnosis (MCID 10 per 1000) assessed with: biopsy and magnetic resonance imaging follow-up: median 3.2 years	8 per 1000	**14 per 1000** (12 to 17)	**RR 1.85** (1.56 to 2.19)	60745 (1 RCT)	⊕⊕⊕⊖ Moderate[a]	Screening likely results in little to no difference in prostate cancer diagnoses.
Localised prostate cancer diagnosis (T1-T2 N0/Nx M0/Mx) (MCID 10 per 1000) assessed with: biopsy and magnetic resonance imaging follow-up: 3.2 years	6 per 1000	**11 per 1000** (9 to 14)	**RR 1.88** (1.55 to 2.27)	60745 (1 RCT)	⊕⊕⊕⊖ Moderate[a]	Screening likely results in little to no difference in localised prostate cancer diagnoses.
Advanced prostate cancer diagnosis (T3–T4 N1 M0/MX, TX NX M1) (MCID 10 per 1000)	1 per 1000	**3 per 1000** (2 to 4)	**RR 1.78** (1.19 to 2.66)	60745 (1 RCT)	⊕⊕⊕⊖ Moderate[a]	Screening likely results in little to no difference in advanced

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assessed with: biopsy and magnetic resonance imaging follow-up: 3.2 years						prostate cancer diagnoses.
Metastatic prostate cancer diagnosis - not reported	-	-	-	-	-	

*__The risk in the intervention group__ (and its 95% confidence interval) is based on the assumed risk in the comparison group and the __relative effect__ of the intervention (and its 95% CI).

CI: confidence interval; **RR:** risk ratio

GRADE Working Group grades of evidence
High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.
Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_457215656520981333.

[a] Downgraded one level due to risk of bias: the included study was at a high risk of bias.

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BACKGROUND

Description of the condition

Prostate cancer is one of the most common malignancies. According to estimates from the International Agency for Research on Cancer's GLOBOCAN database – an internationally recognised source of global cancer statistics – there were approximately 1.5 million new cases of prostate cancer and 396,000 deaths worldwide in 2022 [1]. This makes it the second most commonly diagnosed cancer and the fifth leading cause of cancer death amongst men worldwide [1].

Prostate cancer incidence and mortality differ substantially by geographic region [2]. Northern Europe has the highest estimated incidence rate (82.8 per 100,000 men); other high incidence rate regions are Australia/New Zealand, North America, and Latin America/Caribbean. Southern Africa has the highest mortality rate (29.7 per 100,000 men). South-Central Asia has the lowest rates for both incidence (6.4 per 100,000 men) and mortality (3.1 per 100,000 men). In other words, incidence is about 13 times higher and mortality about 9.5 times higher in the highest-rate regions than in the lowest-rate regions.

The best established risk factors for prostate cancer are increasing age (with most diagnoses in men over 50), a family history in first-degree relatives [3], and ethnicity (with a higher risk in Black men) [4, 5]. Genetic factors are also increasingly recognised. Mutations in BRCA genes are associated with increased risk and have led to the development of targeted treatment approaches [6]. Other genetic variants linked to increased risk include HOXB13 G84E and mutations in the ATM gene [7]. In addition, common single-nucleotide polymorphisms (SNPs) across multiple chromosomal regions are each associated with small increases in risk [8]; when combined into a polygenic risk score, they may help identify individuals at higher risk who could benefit from early detection strategies [9].

Prostate cancer incidence has varied considerably over time, mainly related to the degree to which prostate-specific antigen (PSA) testing has been implemented [10]. Most men diagnosed with prostate cancer today are asymptomatic and are identified on the basis of an elevated PSA level. Uncommon presentations of prostate cancer are haematuria, problems emptying the bladder, or bone pain due to metastatic disease [11]. Asymptomatic prostate cancer can also be suspected based on findings of a digital rectal examination [12]. Diagnosis of prostate cancer ultimately requires a prostate biopsy that is most commonly performed as a transrectal ultrasound-guided prostate biopsy. An alternative approach, the transperineal biopsy, avoids the rectum and is increasingly used because it substantially reduces the risk of infection [13]. Magnetic resonance imaging (MRI) is now commonly used to stratify patient risk and guide biopsy [14]. Well-designed studies support MRI's role in identifying men with clinically relevant prostate cancer while avoiding unnecessary biopsies [15, 16, 17, 18].

The natural history of asymptomatic, screen-detected prostate cancer patients varies greatly. Most of these cancers are clinically localised at the time of diagnosis and have a protracted, indolent course, but a subset will progress or present as metastatic disease at the outset. One of the most powerful predictors of disease course is the histological cancer grade, typically reported using the Gleason scoring system [19]. Since 2016, it has gradually been replaced by the International Society of Urological Pathology (ISUP) system [20]. Other important prognostic factors include clinical stage and PSA level, particularly when combined into established risk classification systems, such as the D'Amico system, the National Comprehensive Cancer Network (NCCN) groups, or the Cancer of the Prostate Risk Assessment (CAPRA) score, which help predict disease progression and guide treatment decisions [21]. Men with very low risk, and even some with intermediate-risk disease, tend to have a very protracted and indolent disease. Men with high-risk disease are more likely to harbour micro-metastatic disease (metastases that are too small to be detected by current staging studies), develop distant metastases to the lymph nodes, bones and other locations, and may develop local complications such as haematuria, retention, and ureteral obstruction. Treatment is tailored accordingly, with guidelines recommending close monitoring and deferred treatment in men at low risk of progression, watchful waiting without curative intent in those individuals unlikely to experience any disease-related morbidity during their lifetime, and systemic treatment in men with metastatic disease [4, 5, 22, 23]. Local treatment with curative intent in the form of radical surgery or radiation treatment is best reserved for men with clinically localised disease that is likely to progress and who have an estimated life expectancy of 10 to 15 years.

Ideally, a prostate cancer screening test would preferentially identify aggressive prostate cancers that are most likely to harm men during their lifespan, rather than finding indolent cancers that may never cause any problems [24, 25, 26, 27].

Description of the intervention and how it might work

The PSA test is a blood test used as a primary screening tool to detect prostate cancer at an early stage. The rationale for early detection is to provide treatment with curative intent, and to reduce overall and disease-specific mortality [28]; ideally, without compromising quality of life or causing adverse events.

PSA is a serine protease secreted exclusively by prostatic epithelial cells into the ejaculate. However, small quantities can also be measured in the blood serum. There is a strong correlation between benign prostatic enlargement and PSA levels [29]. However, levels are also frequently elevated in men with prostate cancer, with levels correlating with tumour volume [30]. While measurement of serum PSA is widely accepted as a sensitive marker for monitoring the progression of established prostate cancer and response to therapy, its utility in early detection of prostate cancer is more controversial.

Prostate cancer is suspected when PSA levels are elevated (although no firm threshold has been established and men with low PSA levels may harbour prostate cancer [31]) or rise quickly over time. Traditionally, further diagnostic work-up consists of an ultrasound-guided biopsy. More recently, magnetic resonance imaging has been adopted as an additional tool to determine prostate cancer risk, better target the biopsy, or both [14, 32].

Why it is important to do this review

Prostate cancer is a leading cause of morbidity and mortality amongst men worldwide. An effective screening programme would identify prostate cancer cases at earlier stages when patients can be cured. The goal of any screening program is to reduce disease-

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specific and overall mortality, in addition to improving a person's future quality of life. However, PSA testing also has potential adverse effects. These include psychological stress, false-positive and false-negative test results, the inconvenience and burden of invasive follow-up tests (such as prostate biopsy), potential complications of such invasive testing (e.g. sepsis), unnecessary treatment of indolent disease, and the downstream complications of prostate cancer treatment, such as bowel and/or bladder incontinence and erectile dysfunction [33].

Several trials have evaluated the effectiveness of screening for prostate cancer compared with no screening [34, 35, 36, 37, 38, 39]. While some trials suggest an overall benefit of prostate cancer screening, others do not. In order to inform individualised decision-making and health policy decisions, the entire body of evidence needs to be considered [40]. This can be best addressed with a systematic review and meta-analysis of randomised controlled trials that compare the effectiveness of screening to no-screening.

In 2006, the initial version of this Cochrane review demonstrated insufficient evidence to either support or refute the use of mass, selective, or opportunistic screening for prostate cancer [41]. The updated Cochrane review published in 2013 included three additional trials for a total of five trials with 341,342 participants [33]. Pooled estimates demonstrated that screening led to increased detection of prostate cancer cases but did not affect prostate cancer-specific mortality nor all-cause mortality [33, 42].

Since 2013, several included trials have provided extended follow-up data [36, 37, 38]. In addition, two new trials have been published [39, 43], providing the rationale to update this review. Meanwhile, conflicting recommendations by major guideline developers – including the United States Preventive Services Task Force (USPSTF) [44], American Cancer Society [45], American Urologic Association (AUA) [46], the UK's National Institute for Health and Care Excellence [47], and the European Urological Association [4] – reflect the ongoing controversy surrounding the relative merits of PSA screening.

The purpose of this review is to determine the effects of prostate cancer screening by updating prior Cochrane reviews using up-to-date methodological standards and including new and extended follow-up trial data. The results of this review can then be used to inform evidence-based individual decision-making at the point of care, serve as the basis of clinical practice guideline recommendations, and help inform healthcare policy.

OBJECTIVES

To assess the effects of prostate cancer screening compared to no screening in men with no prior diagnosis of prostate cancer.

METHODS

We followed the Methodological Expectations of Cochrane Intervention Reviews (MECIR) when conducting the review and PRISMA 2020 for the reporting [48, 49, 50]. The protocol underlying this review was published in 2004 [51], and therefore predates many current methodological requirements of Cochrane reviews, such as using GRADE to assess the certainty of evidence by outcome.

In this update, we excluded quasi-randomised controlled trials since they are at an increased risk of bias. This led to the exclusion of the Norrkoping study [35], excluded due to its pseudo-RCT design.

In the original review protocol, outcomes were less precisely defined and did not consistently reflect patient-important outcomes. In accordance with Cochrane standards, we now focus on outcomes that are directly important to men. We did not include false-positive or false-negative test results as critically important outcomes in this review. We excluded these measures because they represent intermediate test characteristics rather than downstream patient-relevant clinical outcomes, and their interpretation is highly dependent on diagnostic thresholds, follow-up strategies, and definitions used across studies. Following the editorial policy of the Cochrane Urology Group, we also did not formally include costs, which were also listed as an outcome in the original protocol. All current review outcomes are included in the summary of findings tables.

We changed the title to acknowledge PSA-based screening as the focus of this review rather than other forms of screening with digital rectal examination, MRI, or prostate-specific membrane antigen positron emission tomography (PSMA PET) imaging, which fall outside the scope of this review.

Criteria for considering studies for this review

Types of studies

We included all randomised controlled trials (RCTs) comparing PSA-based prostate cancer screening to no screening. We also included cluster-randomised trials. We excluded cross-over trials as these study designs are not relevant in this setting. In addition, we excluded pseudo-randomised trials given their high risk of selection bias. We did not impose language restrictions, and we considered both published and unpublished sources.

Types of participants

All men enrolled in studies of prostate cancer screening were eligible for this review, with no exclusions based on ethnicity, age, or presence of lower urinary tract symptoms. We excluded studies that enrolled men with a previous diagnosis and/or treatment of prostate cancer.

Types of interventions

We defined screening as the use of the prostate-specific antigen (PSA) test, or the PSA test in combination with digital rectal examination (DRE) or other adjuvant markers (e.g. kallikrein).

We included the comparison of PSA-based screening versus no screening. We did not include studies comparing downstream diagnostics or treatment options following an initial PSA-based screening.

Outcome measures

We did not use the measurement of the outcomes assessed in this review as an eligibility criterion.

Critical outcomes

- Prostate cancer-specific mortality: measured as the number of deaths due to prostate cancer [52]. We defined the minimal

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clinically important difference (MCID) as 0.1% (or an absolute difference of 1 in 1000 men).

- Overall mortality: measured as the number of deaths due to any cause [52]. We defined the MCID as 0.1% (or an absolute difference of 1 in 1000 men).
- Adverse events: measured as the number of participants with adverse events, including the downstream consequences (e.g. diagnostic and treatment cascade). We only found data on mortality related to diagnostic biopsy or prostate cancer treatment (CAP [53, 54, 55, 56, 57, 58]). We defined the MCID as 0.1% (or an absolute difference of 1 in 1000 men).

Important outcomes

- Quality of life: final value or change from baseline assessed with validated questionnaires such as the UCLA–Prostate Cancer Index (UCLA-PCI [59]) and the European Organisation for Research and Treatment of Cancer core quality of life questionnaire (EORTC QLQ C-30 [60]). We defined the MCIDs in adverse effects of screening (compared to no screening) as 8, 7, 8, 9, 8, and 11 points higher on the UCLA-PCI scales measuring urinary function, bowel function, sexual function, urinary bother, bowel bother, and sexual bother, respectively [61]. If available, we considered MCIDs of the EORTC QLQ C-30 based on previous literature [62]. We also considered generic quality of life scores such as the 36-Item Short Form Health Survey (SF-36). We considered an MCID to be 9 to 10 points (0.09 to 0.01 in SF-6D utility values), which represents the average MCID across the domains of the instrument [61], which also translates to other generic scales, such as the 15D health state description system, and the EQ-5D instrument [63].
- Prostate cancer diagnosis: measured as the number of prostate cancer cases. We defined the MCID as 1.0% (or an absolute difference of 10 in 1000 men).
- Localised (stage I and stage II) prostate cancer diagnosis. We defined the MCID as 1.0% (or an absolute difference of 10 in 1000 men).
- Advanced (stage III and IV) prostate cancer diagnosis. We defined the MCID as 1.0% (or an absolute difference of 10 in 1000 men).
- Metastatic prostate cancer diagnosis: measured as the number of men diagnosed with metastatic disease due to prostate cancer. We defined the MCID as 0.5% (or an absolute difference of 5 in 1000 men).

We categorised new cases of prostate cancer at the time of diagnosis as localised, advanced, and metastatic, according to the European Association of Urology (EAU) risk categories [64]. To rate the certainty of evidence, we calculated the clinically important differences for each outcome of interest [65, 66]. We defined a clinically meaningful benefit from screening as a mean difference (MD) or an absolute risk difference (aRD) greater than or equal to the minimal clinically important difference (MCID). Since we did not find any published information on what constitutes a clinically important reduction, these thresholds were informed by the content expertise of the author team (JF, ECH, JHJ, and PD).

Search methods for identification of studies

We performed a comprehensive search without language or publication status restrictions.

Electronic searches

We expanded the search strategies used in previous systematic reviews on prostate cancer screening by searching additional databases and revising the search terms to reflect changes in medical subject heading terminology. Since these changes were substantial, we searched all databases back to their inception. Our last systematic search was on 19 November 2025. See Supplementary material 1 for the full search strategies. We searched the following databases.

- Cochrane Library Issue 11 2025 (via Wiley)
  - Cochrane Database of Systematic Reviews (CDSR)
  - Cochrane Central Register of Controlled Trials (CENTRAL)
  - Database of Abstracts of Reviews of Effects (DARE) (Up to 7 August 2018)
  - Health Technology Assessment Database (HTA) (Up to 7 August 2018)
- MEDLINE (via OvidSP): 1946 onwards
- MEDLINE In-Process & Epub Ahead of Print (via OvidSP): searched 19 November 2025
- PubMed (via NCBI): latest year only (19 November 2024 to 19 November 2025) to capture epubs ahead of print not yet in MEDLINE
- Embase (via OvidSP): 1947 onwards
- LILACS (Latin American and the Caribbean Health Sciences Literature; via VHL): 1982 onwards
- Scopus (via Elsevier): 1960 onwards
- Web of Science Core Collection (via Thomson Reuters): from 1900 up to 1 December 2025
- OpenGrey (via INIST-CNRS): 1980 to 2020 (discontinued)

We also searched the following trial registers up to 19 November 2025.

- ClinicalTrials.gov (https://clinicaltrials.gov/)
- World Health Organization International Clinical Trials Registry Platform search portal (https://trialsearch.who.int/Default.aspx)

Searching other resources

We searched the reference lists of the included trials and any ancillary publications (reviews, meta-analyses, and health technology assessment reports). We contacted the authors of the included trials to identify any additional studies that we might have missed. We handsearched the abstract proceedings of the annual meetings of the American Urological Association, European Association of Urology, and the American Society of Clinical Oncology to identify unpublished studies between 2014 and 2018. More recent years (2019 to 2025) were included in electronic databases and captured through the electronic search strategy. We also conducted a search for post-publication amendments and retractions in MEDLINE (OvidSP) and Embase (OvidSP) to ensure none of the included studies have been withdrawn or include errors. We conducted the most recent search on 19 November 2025.

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Data collection and analysis

Selection of studies

We used the reference management software EndNote to identify and remove potential duplicate records [67]. Two authors (ECH and JF) independently reviewed all references to determine eligibility, in accordance with guidance in the *Cochrane Handbook for Systematic Reviews of Interventions* [68]. We used Covidence for title/abstract and full-text screening (and further de-duplication) [69]. We resolved any discrepancies through consensus by consulting a third review author (PD). For excluded studies that may have reasonably been expected to be included in this review, we documented our reasons for exclusion in the 'Characteristics of excluded studies' table (see Supplementary material 3). We report our process for study selection in an adapted PRISMA flow diagram [70].

Data extraction and management

We used a dedicated extraction form to extract outcomes data to calculate summary statistics and measures of variance. For studies that met inclusion criteria, two of five review authors (DI, JF, ECH, OV, and JHJ) independently extracted the following information, which we report in the 'Characteristics of included studies' table (see Supplementary material 2).

- Study design
- Study dates (if dates were unavailable, we reported this)
- Study settings and country
- Participant inclusion and exclusion criteria
- Participant details and baseline demographics (such as age)
- Number of participants in each study and study arm
- Details of relevant comparator and experimental interventions (e.g. screening interval, biopsy thresholds)
- Definitions of relevant outcomes, method and timing of outcome measurement, and definitions of any relevant subgroups
- Rates of adherence in the intervention group and contamination in the control group, respectively
- Study funding sources
- Primary investigators' declarations of interest

For dichotomous outcomes, we populated a 2x2 table with the number of events and total number of participants randomised to each study arm, as well as each outcome's summary statistics and corresponding measures of variance.

We resolved any disagreements through discussion, or, if required, by consulting a third review author (PD).

We provide information, including trial identifiers, about potentially relevant ongoing studies in the 'Characteristics of ongoing studies' table (see Supplementary material 4).

We contacted the authors of the included studies to obtain key missing data as needed.

Dealing with duplicate and companion publications

For duplicate publications, companion documents, or multiple reports of a primary study, we mapped all publications to a unique study identifier and collated all available data across all known publications. We prioritised publications that reported the longest duration of follow-up.

Risk of bias assessment in included studies

Two review authors (JHJ, ECH, or PD) independently assessed the risk of bias of each included study. We resolved disagreements by consensus, or by consulting a third review author (DI).

We assessed the risk of bias using the Cochrane Collaboration's original tool [71]. We assessed the following domains:

- random sequence generation (selection bias);
- allocation concealment (selection bias);
- blinding of participants and personnel (performance bias);
- blinding of outcome assessment (detection bias);
- incomplete outcome data (attrition bias);
- selective reporting (reporting bias);
- adherence;
- contamination;
- other sources of bias.

We judged the risk of bias domains as 'low risk', 'high risk', or 'unclear risk', and evaluated individual bias items as described in the *Cochrane Handbook* [71].

For performance bias (blinding of participants and personnel), we considered all outcomes similarly susceptible to performance bias.

Regarding the risk of detection bias, we defined all outcomes but overall mortality as subjective outcomes since they involved some judgement on the part of the outcome assessor, who was either the participant himself or a member of the investigative team.

We assessed attrition bias (incomplete outcome data) on an outcome-specific basis but collapsed reporting of outcomes with identical judgements. For reporting bias (selective reporting), we evaluated whether each trial's prespecified protocol (when available) matched the outcomes and analysis described in the reported trial.

For each domain, we first assessed the outcomes reported in each study independently. We then summarised the risk of bias of each outcome across all studies. This approach follows the risk of bias assessment detailed in the sixth edition of the *Cochrane Handbook* [72].

Measures of treatment effect

We report dichotomous data as incidence rate ratios (IRRs) when outcomes were reported as person-years; otherwise, we report relative risk ratios (RRs) with 95% confidence intervals (CIs). We express continuous data as mean differences (MDs) with 95% CIs or standardised MDs (SMDs) with 95% CIs.

Unit of analysis issues

The unit of analysis was the individual participant. However, one included trial was cluster-randomised, which we handled according to guidance provided in the *Cochrane Handbook* [73]. Specifically, for the CAP study, we used an adjusted incidence rate ratio derived from a generalised regression approach to account for clustering.

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Dealing with missing data

We contacted the study authors to obtain details about relevant missing data. When feasible, we performed intention-to-treat analyses; otherwise, we performed case analyses. We investigated attrition rates (e.g. dropouts, losses to follow-up, and withdrawals), which informed our assessment of attrition bias. We did not impute missing data.

Reporting bias assessment

We attempted to obtain study protocols to assess for selective outcome reporting. We did not generate funnel plots to assess small-study effects because all analyses included fewer than 10 studies [33].

Synthesis methods

We summarised the data using a random-effects model as it provides the most conservative pooled effect size estimate. We performed all statistical analyses according to statistical guidelines in the *Cochrane Handbook for Systematic Reviews of Interventions* [68]. For dichotomous outcomes, we used the Mantel-Haenszel method. For continuous outcomes, we used the inverse-variance method. We used Review Manager software to perform all analyses [74]. We used adjusted IRRs to account for clustering effects in the CAP trial [75].

We identified heterogeneity (inconsistency) by visually inspecting the degree of CI overlap on each forest plot. We also calculated the I^2 statistic, which quantifies inconsistency across studies to assess the impact of heterogeneity on the meta-analysis [76, 77]. We interpreted I^2 as follows [78]:

- 0% to 40%: may not be important;
- 30% to 60%: may indicate moderate heterogeneity;
- 50% to 90%: may indicate substantial heterogeneity;
- 75% to 100%: considerable heterogeneity.

When we identified heterogeneity, we explored potential reasons by examining individual studies and subgroup characteristics.

Investigation of heterogeneity and subgroup analysis

We assumed that the following characteristics may introduce clinical heterogeneity.

- Participant age (younger than 65 years versus 65 years or older). The rationale for this subgroup analysis is that the prevalence of prostate cancer increases with age and the effects of screening may also vary with age [79, 80].
- Screening interval (annual screening versus biannual screening versus one-time screening) since more frequent screening may conceivably increase prostate detection but also result in more adverse events.
- The European Randomized Study of Screening for Prostate Cancer (ERSPC) trial is an international trial comprised of multiple sites. The results of the ERSPC [81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111] trial have mostly been published as individual trials from each site. We conducted our primary analysis using the outcomes report from the entire international cohort. Given the clinical and methodological heterogeneity across national sites, we also performed a secondary subgroup analysis using individual site data (when available), considering each site a unique and separate trial.
- Since the ERSPC investigators conducted their primary analysis on the core age group of men aged 55 to 69 years, we did the same. However, we also performed a secondary subgroup analysis that included the entire cohort of randomised participants.

We limited subgroup analyses to the critical outcomes only, and interpretation included the test for interaction.

Equity-related assessment

No equity-related assessments were planned or conducted in this review.

Sensitivity analysis

We had planned to perform a sensitivity analysis of our critical outcomes limited to those studies judged to be at lower risk of bias. As none of the studies were at low risk of bias, we conducted a sensitivity analysis restricted to the studies (in this case: one) at lower risk of bias (the ERSPC trial).

Certainty of the evidence assessment

We presented the overall certainty of the evidence for each outcome according to the GRADE approach, which takes into account five criteria that measure both internal validity (risk of bias, inconsistency, imprecision, publication bias) and external validity (such as directness of results) [112]. For each comparison, two review authors (DI, JD, JHJ, or ECH) independently rated the certainty of the evidence for each outcome as 'high', 'moderate', 'low', or 'very low' using GRADEpro GDT software [113]. We resolved discrepancies by consensus; if this was not possible, a third review author (PD) arbitrated. We summarised the evidence for the critical and important outcomes (see Outcome measures) in the summary of findings tables, including the numbers of participants and studies addressing each outcome, the best estimate of the relative effect size, and the overall confidence rating for each outcome's effect size estimate [114, 115].

We used what has been referred to as a 'minimally contextualised approach' for rating the certainty of evidence [116], in particular for the domain of imprecision. Based on expert input and benchmarks from other cancer screening programmes, we used a 1 in 1000 (0.1%) minimally important difference for prostate cancer-specific mortality and overall mortality, 10 in 1000 (1%) for prostate cancer diagnosis (including localised and advanced diagnosis), and 5 in 1000 (0.5%) for metastatic prostate cancer diagnosis.

Consumer involvement

We did not have any formal consumer involvement in this review.

RESULTS

Description of studies

Results of the search

The original search (2006) identified 99 potentially relevant articles for full-text review. Of these, two RCTs met the inclusion criteria. The search for our 2010 updated review identified 106 potentially relevant articles, amongst which three new RCTs met the inclusion

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criteria. For our 2013 review update, we retrieved 31 articles for full-text examination [33]. Most were secondary publications of already included studies, with scant additional data. We added these reports to the reference list for completeness.

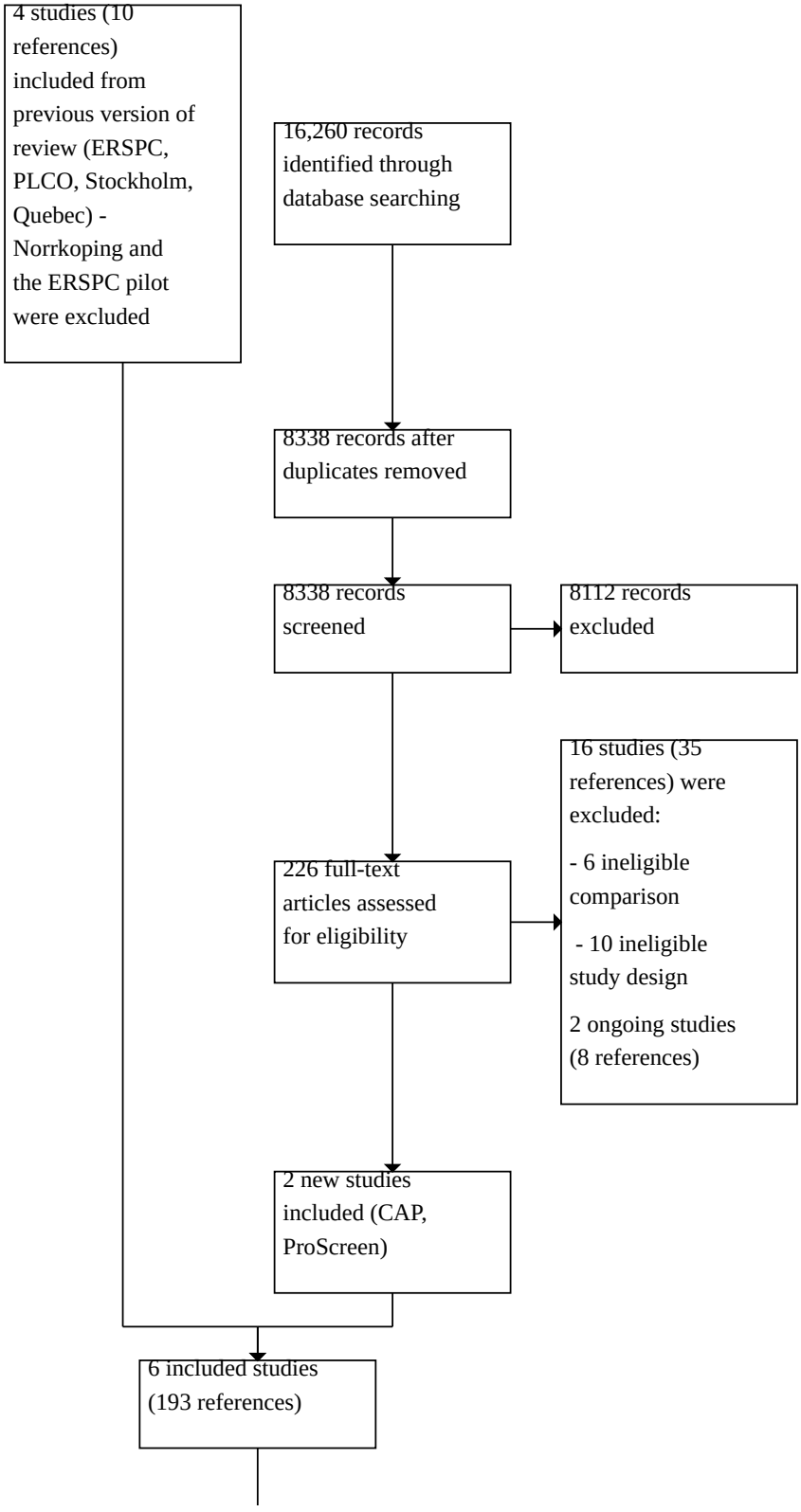
For this update, our most recent search on 19 November 2025 yielded 16,260 citations, from which we selected 226 for full-text review. In total, we excluded 16 studies (35 references), and we identified two ongoing studies (eight references). We identified two new studies (16 reports) in the update searches. Thus, together with four included studies from previous versions, the review now includes a total of six randomised controlled trials (reported in 193 reports).

The studies that we included in this review are: updates on extended follow-up in the Stockholm [117, 118, 119] and ERSPC studies (including data from the individual ERSPC sites: ERSPC (Antwerp) Belgium [120]; ERSPC Finland [121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146]; ERSPC France [147, 148]; ERSPC Italy [149]; ERSPC (Rotterdam) Netherlands [150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180]; ERSPC Spain [181, 182, 183, 184, 185, 186, 187, 188]; ERSPC (Göteborg) Sweden [189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224]; ERSPC (Aargau) Switzerland [225, 226, 227, 228, 229]), as well as the PLCO [230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249, 250, 251, 252, 253, 254, 255, 256, 257, 258, 259] and Quebec [260, 261, 262] studies. Since the study that reported extended follow-up of the entire ERSPC trial (and summarised outcomes at each individual site) did not include ERSPC pilot (Rotterdam) Netherlands 2019 [263, 264, 265, 266, 267, 268, 269], we used the longest duration of follow-up published by this site. We also identified two additional, new studies: the cluster-randomised CAP study and ProScreen [43, 270, 271, 272, 273, 274, 275, 276, 277, 278]. We classified two studies as ongoing (GÖTEBORG-2 2022 [279, 280, 281, 282, 283, 284, 285]; ChiCTR2500102254 [286]).

We describe the systematic search, study selection, and assessment process in a PRISMA diagram (Figure 1).

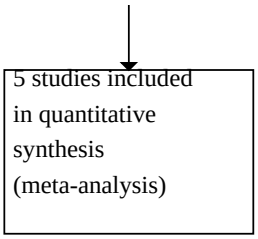
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Figure 1. PRISMA flow diagram



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Figure 1. (Continued)



Included studies

We provide a detailed description of the characteristics of included trials elsewhere (see Supplementary material 2 and Table 1). The following is a succinct overview.

Source of data

We obtained all data presented in this review from published literature, as identified in our electronic search. We contacted the authors of the ERSPC Finland study for quality of life data.

Trial design

Five of the studies included in this review were individual, participant-based RCTs (ERSPC; PLCO; ProScreen; Quebec; Stockholm). One trial, CAP, was a cluster-RCT that randomised all participants assigned to a particular medical trust in the United Kingdom to screening versus no screening.

Settings

Four studies were multicentre trials (CAP; ERSPC; PLCO; ProScreen). The CAP study was conducted across eight centres in the United Kingdom. The ERSPC study was conducted across eight European countries (the Netherlands, Belgium, Sweden, Finland, Italy, Spain, Switzerland, and France). We report findings from the main ERSPC study, as well as the individual ERSPC study sites that have published data exclusively on their cohort of patients (ERSPC (Aargau) Switzerland; ERSPC (Antwerp) Belgium; ERSPC (Göteborg) Sweden; ERSPC (Rotterdam) Netherlands; ERSPC Finland; ERSPC France; ERSPC Italy; ERSPC Spain). The PLCO study was conducted across 10 centres in the USA. The ProScreen study was conducted across two centres in Finland. The remaining studies were single-centre studies: Quebec was conducted in Canada; Stockholm in Sweden.

Overview of trial populations

This review update includes a total of 789,086 randomised participants (355,972 men randomised to screening and 433,114 randomised to the control arm), of which 782,456 (99.1%) were followed up (349,348 in the screening arm and 433,108 in the control arm). The number of enrolled participants ranged from 27,146 (Stockholm) to 415,357 (CAP) participants.

Participants

The details of the baseline characteristics of all trial participants are reported in Table 1 and Table 2. Participants in the included studies were men aged 45 to 80 years. Across all included studies, the median age of participants allocated to screening ranged from 56.9 to 63.2 years, whereas the median age of participants allocated to control groups ranged from 57.1 to 63.2 years. The ERSPC study reported findings across two age cohorts – the 'core' age group, which was defined as men 55 to 69 years of age, and the entire cohort (men aged 50 to 74 years). We conducted our meta-analysis using data from both 'core' and all age groups. All studies excluded men with a previous history of prostate cancer.

Interventions and comparisons

All studies compared screening for prostate cancer (either PSA alone, or in combination with DRE) versus no screening. The included studies varied considerably with respect to the screening frequency, screening interval, and cutoff used to signify a positive PSA screen. Screening protocols ranged from one-time screening to multiple screening rounds and ranged from screening annually to every six years.

- The CAP study compared one-time screening with PSA to no screening. A PSA threshold of ≥ 3 ng/mL prompted confirmatory testing with transrectal ultrasound (TRUS)-guided prostate biopsy.
- The ERSPC study compared screening with PSA (with or without DRE) to no screening. The individual study sites' use of PSA alone or in combination with DRE varied considerably, as did the screening interval (every two to four years). The PSA threshold to indicate the need for further testing varied across sites (ranging from 2.5 to 10 ng/mL).
- The Quebec study compared annual screening with PSA and DRE to no screening. A PSA threshold of ≥ 3 ng/mL prompted confirmatory testing with TRUS-guided prostate biopsy.
- The Stockholm study compared one-time screening with PSA to no screening. A PSA threshold of ≥ 10 ng/mL prompted further investigation by TRUS-guided prostate biopsy.
- The PLCO study compared annual screening with PSA and DRE to no screening. A PSA threshold of ≥ 4 ng/mL or a DRE suspicious of cancer prompted further investigation. The PLCO study did not oversee the diagnostic process.
- The ProScreen study compared one-time screening with PSA to no screening. A PSA threshold of ≥ 3 ng/mL prompted reflex testing with a kallikrein panel, and if the panel indicated elevated risk, an MRI was performed.

The duration of participant follow-up ranged from 3.2 to 23 years across the trials.

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Outcomes

All studies reported either overall or prostate cancer-specific mortality as their primary endpoint. Adverse events related to screening, specifically complications of biopsy, were reported only in the CAP and PLCO studies. Only the Finnish arm of the ERSPC reported data on quality of life when comparing screening to no screening (ERSPC Finland). The trials used independent evaluation committees (who were blinded to trial group assignment), national death registries, and individual medical records (including death certificates) to confirm these outcomes. Please refer to the Synthesis of results section for specific details on all outcomes.

Excluded studies

We excluded 16 studies after full-text review of each publication because they did not meet the review's inclusion criteria. They were primarily studies comparing downstream diagnostic or treatment approaches following baseline screening with PSA. We report further details in Supplementary material 3. Of note, although we included the Norrkoping study in earlier versions of this review (Sandblom 2011 (Norrköping study) [287, 288, 289, 290, 291, 292, 293]), we excluded it from this review because of its pseudo-RCT design. We also excluded the ERSPC pilot study (ERSPC pilot (Rotterdam) Netherlands 2019), included in the previous update, as it considered a first round of PSA screening for all participants and then subsequent testing.

Ongoing studies

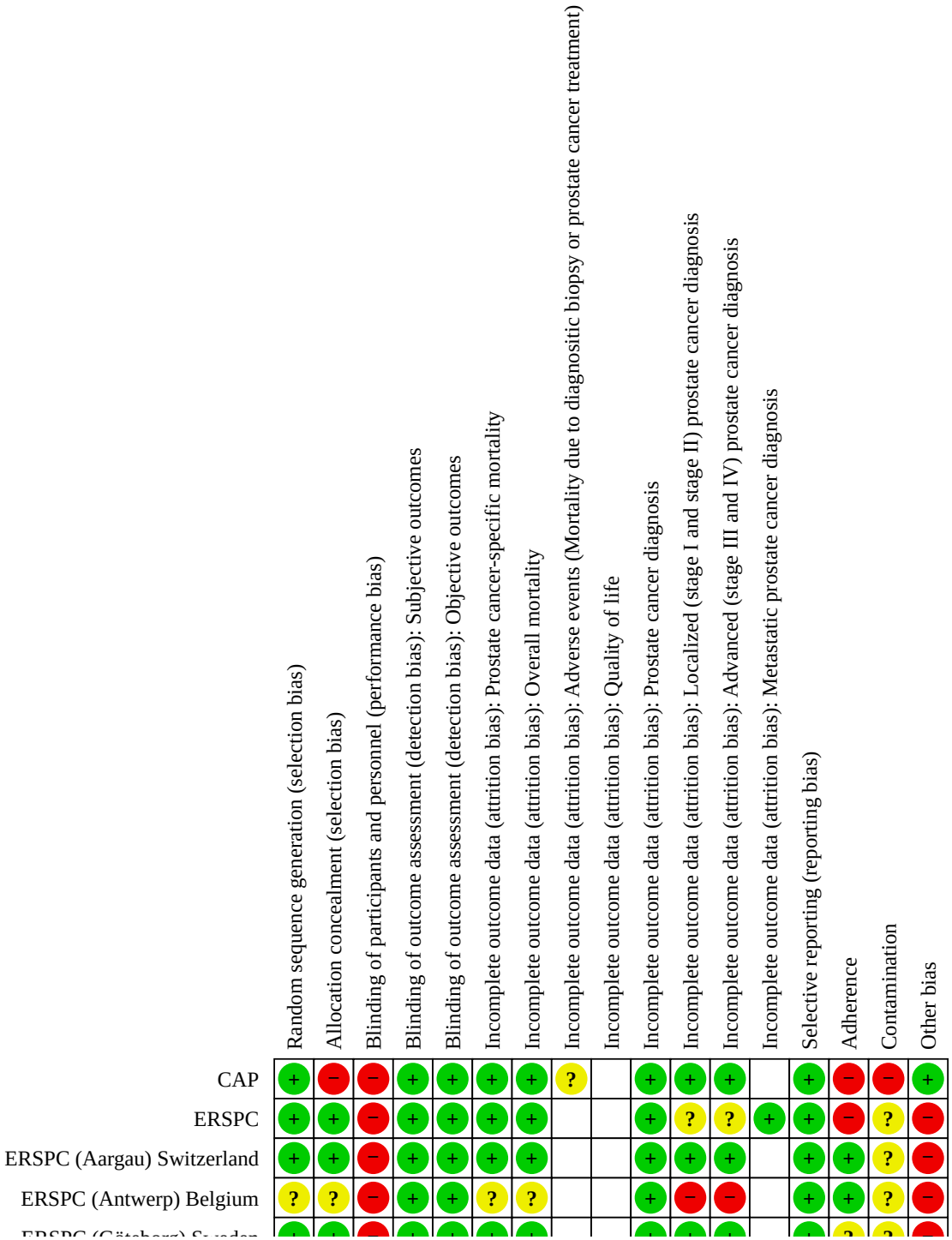
The GÖTEBORG Prostate Cancer Screening 2 trial (G2) is a population-based screening study recruiting men aged 50 to 60 from West Sweden and randomising them to screening or control (GÖTEBORG-2 2022). Trial accrual started in September 2015 and is estimated to conclude in December 2040. We identified publications from this study in relation to the first screening round and other subsequent analyses, but not for this review's comparison of interest. Another trial from China randomising men to PSA screening or control was registered in 2025 (ChiCTR2500102254). See details in Supplementary material 4.

Risk of bias in included studies

For full details on the risk of bias of the included trials, please refer to Supplementary material 2 and Figure 2.

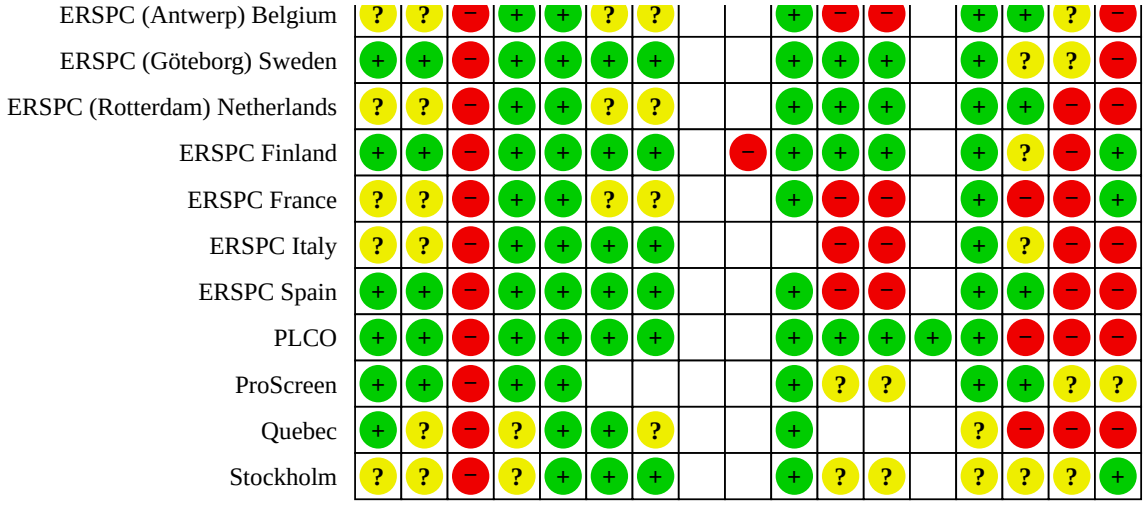
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Figure 2. Risk of bias summary: review authors' judgements about each risk of bias item for each included trial (blank cells indicate that the trial did not measure that particular outcome).



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Figure 2. (Continued)



Random sequence generation (selection bias)

We judged the random sequence generation to be at low risk of bias in most included trials, including CAP, PLCO, ProScreen, Quebec, and most ERSPC centres. For several ERSPC centres, the method of sequence generation was insufficiently described, resulting in an unclear risk of bias.

Allocation concealment (selection bias)

Allocation concealment was generally adequately reported and judged at low risk of bias in ProScreen, PLCO, and most ERSPC centres. However, for several ERSPC centres, details on allocation concealment were lacking, leading to an unclear risk of bias. We assessed CAP as high risk of bias because there was no allocation concealment, and participants consented to trial enrolment after randomisation.

Blinding of participants and personnel (performance bias)

Blinding of participants and personnel was not feasible due to the nature of the screening interventions. Consequently, we judged all trials to be at high risk of performance bias.

Blinding of outcome assessment (detection bias)

We considered blinding of outcome assessment to be at low risk of bias for most subjective and objective outcomes, including prostate cancer-specific mortality and overall mortality, respectively. For outcomes requiring clinical judgement, reporting was insufficient, resulting in unclear risk of bias in two trials (Quebec; Stockholm).

Incomplete outcome data (attrition bias)

For mortality outcomes, incomplete outcome data were generally judged to be at low risk of bias across trials, reflecting comprehensive follow-up through cancer and mortality registries. For non-mortality outcomes, including prostate cancer diagnosis and disease stage at diagnosis, attrition bias was more variable. Several trials showed an unclear or high risk of bias due to incomplete reporting of losses to follow-up or differential ascertainment between groups. Adverse events related to diagnostic biopsy or prostate cancer treatment were frequently inadequately reported, leading to an unclear risk of attrition bias for this outcome (CAP). Furthermore, quality of life was measured only in a subset of participants in the ERSPC Finland study, so we considered the risk of bias to be high.

Selective reporting (reporting bias)

We judged two trials to have an unclear risk of selective reporting bias (Quebec; Stockholm).

Adherence

We judged the CAP, ERSPC, ERSPC France, PLCO, and Quebec studies to have a high risk of bias, due to low adherence rates. We judged other studies as having an unclear or low risk of bias based on their description of adherence data.

Contamination

We considered contamination of the control group through opportunistic PSA testing likely in several trials, including the CAP, PLCO, ERSPC (and many of the sites), and Quebec studies. For other sites or studies, contamination was not reported.

Other bias

We identified no other sources of bias for the CAP and Stockholm studies. We identified changes in PSA threshold amongst sites as a potential source of bias for the ERSPC study. Overall mortality may have been impacted by colorectal cancer screening for the PLCO study. We judged the Quebec study to have a high risk of bias as analysis was not performed according to the intention-to-treat principle.

Overall assessment

Overall, we judged only the ERSPC study as having a lower risk of bias. We assessed three domains as high risk of bias: blinding of participants and personnel; adherence; and other bias. However, it should be recognised that we assessed all studies as having a high risk of bias for blinding of participants and personnel. We determined that the source of bias in the remaining two domains did not affect the primary outcome of the ERSPC study. The

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remaining studies had an overall moderate or high risk of bias and the source of bias possibly affected the primary outcome of interest in each study.

Synthesis of results

Comparison 1. Screening based on PSA (alone) – all studies

See Summary of findings 1. These results include all studies, including the core age group for the ERSPC study.

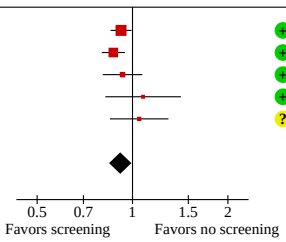
Critical outcomes

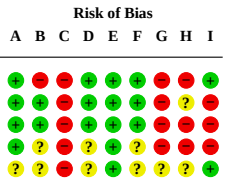
Prostate cancer-specific mortality

Screening may reduce prostate cancer-specific mortality (Rate ratio 0.91, 95% CI 0.85 to 0.99; I^2 = 0%; 5 studies, 721,607 participants; low-certainty evidence; Figure 3). Assuming a higher baseline risk of 16 prostate cancer-related deaths per 1000 men, as reported in the ERSPC study at 23 years, this translates to 1 fewer death (95% CI 2 fewer to 0 fewer) per 1000 individuals. We downgraded the certainty of the evidence for study limitations and imprecision. This interpretation was based on an MCID of 1 per 1000 men screened.

Figure 3. PSA screening versus no screening: prostate cancer-specific mortality

Study or Subgroup	log[Rate Ratio]	SE	Screening Total	No screening Total	Weight	Rate Ratio IV, Random, 95% CI
CAP[a]	-0.083382	0.038896	189326	219395	43.2%	0.92 [0.85 , 0.99]
ERSPC[b]	-0.1393	0.0428	72888	89348	35.7%	0.87 [0.80 , 0.95]
PLCO[c]	-0.072571	0.07339	38340	38343	12.1%	0.93 [0.81 , 1.07]
Quebec[d]	0.076961	0.140081	31133	15353	3.3%	1.08 [0.82 , 1.42]
Stockholm[e]	0.04879	0.108391	2400	25081	5.6%	1.05 [0.85 , 1.30]
Total (HKSJ[f])			**334087**	**387520**	**100.0%**	**0.91 [0.85 , 0.99]**





Test for overall effect: T = 3.30, df = 4 (P = 0.03)

Heterogeneity: Tau² ($REML_g$) = 0.00; Chi² = 4.47, df = 4 (P = 0.35); I² = 0%

Footnotes

[a]As randomised - 15 Year Follow-Up
[b]As randomised - 23 Year Follow-Up (core ages)
[c]As randomised - 16.9 Year Follow-Up
[d]As randomised - 11 Year Follow-Up
[e]As randomised - 20 Year Follow-Up
[f]CI calculated by Hartung-Knapp-Sidik-Jonkman (HKSJ) method.
[g]Tau² calculated by Restricted Maximum-Likelihood method.

Risk of bias legend

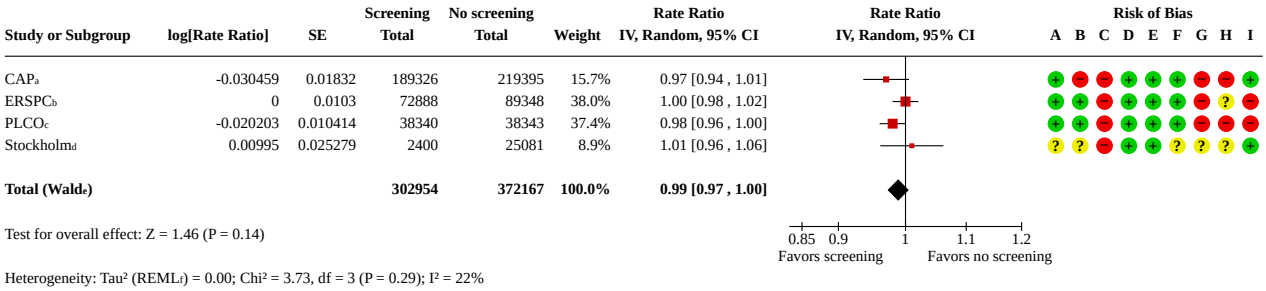
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias): Subjective outcomes
(E) Incomplete outcome data (attrition bias): Prostate cancer-specific mortality
(F) Selective reporting (reporting bias)
(G) Adherence
(H) Contamination
(I) Other bias

Overall mortality

Screening may also reduce overall mortality, although the confidence interval includes the possibility of little to no effect (Rate ratio 0.99, 95% CI 0.97 to 1.00; I^2 = 22%; 4 studies, 675,121 participants; low-certainty evidence; Figure 4). Assuming a higher baseline risk of 491 deaths from any cause per 1000 men, as reported in the ERSPC study at 23 years, this corresponds to 5 fewer deaths (95% CI 15 fewer to 0 fewer) per 1000. We downgraded the certainty of the evidence for study limitations and imprecision. This interpretation was based on an MCID of 1 per 1000 men screened.

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Figure 4. PSA screening versus no screening: overall mortality



Footnotes

[a]As randomised - 15 Year Follow-Up
[b]As randomised - 23 Year Follow-Up (core ages)
[c]As randomised - 16.9 Year Follow-Up
[d]As randomised - 20 Year Follow-Up
[e]CI calculated by Wald-type method.
[f]Tau² calculated by Restricted Maximum-Likelihood method.

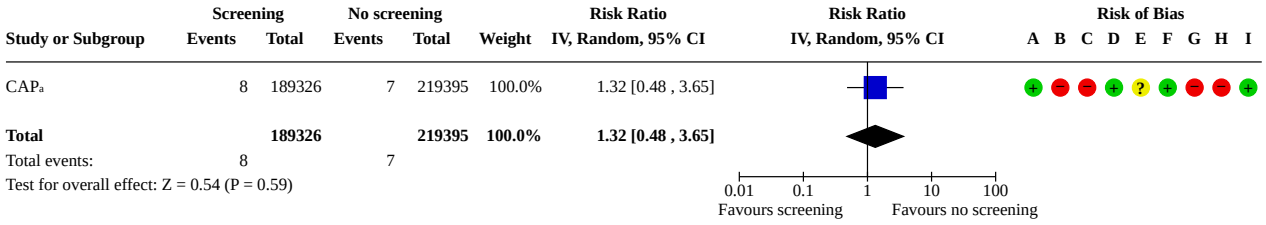
Risk of bias legend

(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias): Objective outcomes
(E) Incomplete outcome data (attrition bias): Overall mortality
(F) Selective reporting (reporting bias)
(G) Adherence
(H) Contamination
(I) Other bias

Adverse events

Screening may not increase adverse events (RR 1.32, 95% CI 0.48 to 3.65; I^2 not applicable; 1 study, 408,721 participants; low-certainty evidence; Figure 5). Assuming a baseline risk of 3 adverse events per 100,000 men screened, as reported in the CAP study, this corresponds to 1 more adverse event per 100,000 men screened (95% CI 2 fewer to 8 more). We downgraded for study limitations and indirectness because the one included study reported only screening-related deaths, but not other adverse events such as hospitalisation due to urosepsis related to prostate biopsies. This interpretation was based on an MCID of 1 per 1000 men screened.

Figure 5. PSA screening versus no screening: adverse events (defined as mortality due to diagnostic biopsy or prostate cancer treatment)



Footnotes

[a]As randomised - 15 Year Follow-Up

Risk of bias legend

(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias): Objective outcomes
(E) Incomplete outcome data (attrition bias): Adverse events (Mortality due to diagnositic biopsy or prostate cancer treatment)
(F) Selective reporting (reporting bias)
(G) Adherence
(H) Contamination
(I) Other bias

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Important outcomes

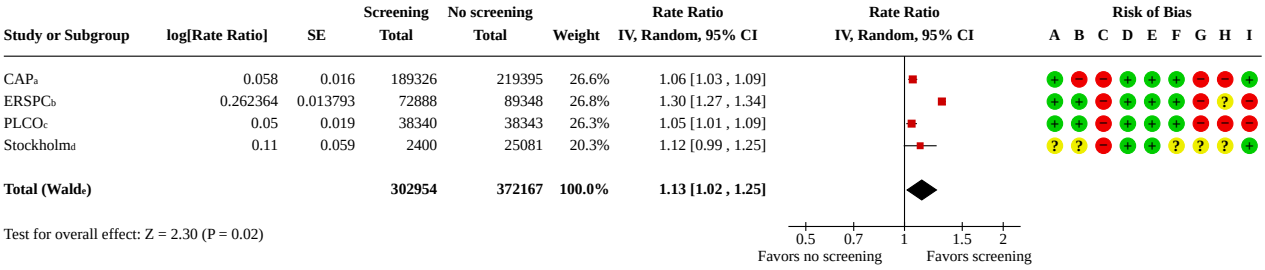
Quality of life

The Finnish arm of the ERSPC trial reported data on quality of life in a random sample of 1100 participants in each arm of the trial (ERSPC Finland). Men diagnosed with prostate cancer were excluded from the analysis. There was little to no difference between the screening and the control group (SF-6D: screening 0.777 (n = 486) versus control 0.779 (n = 483); P = 0.88; MCID 0.09 to 0.10). Results from other scores were similar (15D score: screening 0.890 (n = 534) versus control 0.895 (n = 522); P = 0.62 and EQ-5D score: screening 0.830 (n = 519) versus control 0.857 (n = 514); P = 0.08) [135]. We downgraded the certainty of the evidence twice for study limitations (due to lack of allocation concealment, attrition bias, and the exclusion of men with a diagnosis of prostate cancer from this analysis).

Prostate cancer diagnosis

Screening may increase prostate cancer diagnoses (Rate ratio 1.13, 95% CI 1.02 to 1.25; I^2 = 96%; 4 studies, 675,121 participants; low-certainty evidence; Figure 6). Assuming a baseline risk of 120 prostate cancer diagnoses per 1000, as reported in ERSPC at 23 years, this corresponds to 16 more diagnoses (95% CI 2 more to 30 more) per 1000. We downgraded the certainty of the evidence for study limitations and inconsistency. We did not downgrade further for imprecision, as we attributed the apparent imprecision to inconsistency. This interpretation was based on an MCID of 10 per 1000 men screened.

Figure 6. PSA screening versus no screening: prostate cancer diagnosis



Test for overall effect: Z = 2.30 (P = 0.02)

Heterogeneity: Tau² (REML[f]) = 0.01; Chi² = 127.32, df = 3 (P < 0.00001); I² = 96%

Footnotes

[a]As randomised - 15 Year Follow-Up
[b]As randomised - 23 Year Follow-Up (core ages)
[c]As randomised - 16.9 Year Follow-Up
[d]As randomised - 20 Year Follow-Up
[e]CI calculated by Wald-type method.
[f]Tau² calculated by Restricted Maximum-Likelihood method.

Risk of bias legend

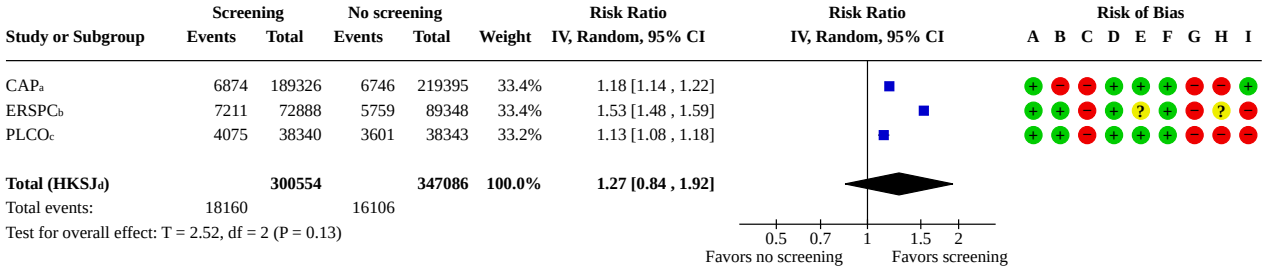
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias): Objective outcomes
(E) Incomplete outcome data (attrition bias): Prostate cancer diagnosis
(F) Selective reporting (reporting bias)
(G) Adherence
(H) Contamination
(I) Other bias

Localised (stage I and II) prostate cancer diagnosis

We are uncertain about the effects of screening on the diagnosis of localised prostate cancer (RR 1.27, 95% CI 0.84 to 1.92; I^2 = 99%; 3 studies, 647,640 participants; very low-certainty evidence; Figure 7). We downgraded the certainty of the evidence for study limitations, inconsistency, and imprecision. This interpretation was based on an MCID of 10 per 1000 men screened.

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Figure 7. PSA screening versus no screening: localised (stage I and II) prostate cancer diagnosis



Heterogeneity: Tau² (REML[e]) = 0.03; Chi² = 169.04, df = 2 (P < 0.00001); I² = 99%

Footnotes

[a]As randomised - 15 Year Follow-Up
[b]As randomised - 23 Year Follow-Up (core ages)
[c]As randomised - 16.9 Year Follow-Up
[d]CI calculated by Hartung-Knapp-Sidik-Jonkman (HKSJ) method.
[e]Tau² calculated by Restricted Maximum-Likelihood method.

Risk of bias legend

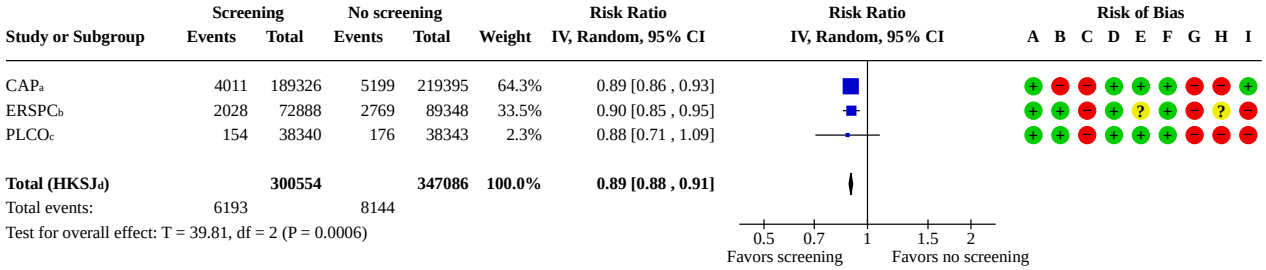
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias): Subjective outcomes
(E) Incomplete outcome data (attrition bias): Localized (stage I and stage II) prostate cancer diagnosis
(F) Selective reporting (reporting bias)
(G) Adherence
(H) Contamination
(I) Other bias

Advanced (stage III and IV) prostate cancer diagnosis

Screening likely results in little to no difference in advanced (stage III and IV) prostate cancer diagnosis (RR 0.89, 95% CI 0.88 to 0.91; I^2 = 0%; 3 studies, 647,640 participants; moderate-certainty evidence; Figure 8). Assuming a baseline risk of 31 advanced prostate cancer diagnoses per 1000, as reported in the ERSPC study at 23 years, this corresponds to 3 fewer diagnoses (95% CI 4 fewer to 3 fewer) per 1000. We downgraded the certainty of the evidence for study limitations only. We did not downgrade for the inconsistency observed because it did not affect the interpretation of the results. This interpretation was based on an MCID of 10 per 1000 men screened.

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Figure 8. PSA screening versus no screening: advanced (stage III and IV) prostate cancer diagnosis



Heterogeneity: Tau² (REML[e]) = 0.00; Chi² = 0.06, df = 2 (P = 0.97); I² = 0%

Footnotes

[a]As randomised - 15 Year Follow-Up
[b]As randomised - 23 Year Follow-Up (core ages)
[c]As randomised - 16.9 Year Follow-Up
[d]CI calculated by Hartung-Knapp-Sidik-Jonkman (HKSJ) method.
[e]Tau² calculated by Restricted Maximum-Likelihood method.

Risk of bias legend

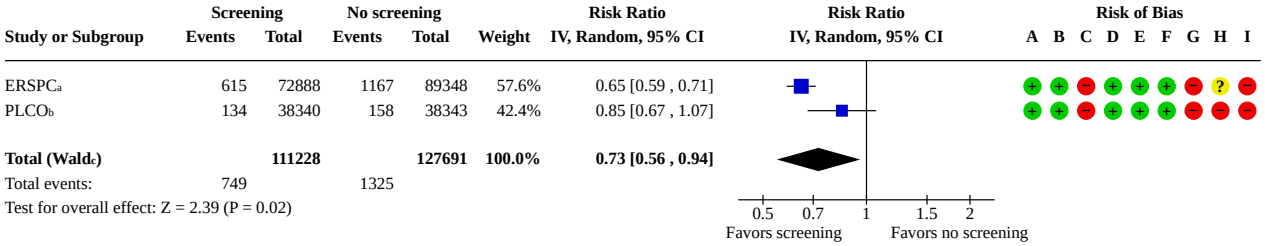
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias): Subjective outcomes
(E) Incomplete outcome data (attrition bias): Advanced (stage III and IV) prostate cancer diagnosis
(F) Selective reporting (reporting bias)
(G) Adherence
(H) Contamination
(I) Other bias

Metastatic prostate cancer diagnosis

Screening may result in little to no difference in the diagnosis of metastatic prostate cancer disease (RR 0.73, 95% CI 0.56 to 0.94; I^2 = 78%; 2 studies, 238,919 participants; low-certainty evidence; Figure 9). Assuming a baseline risk of 13 metastatic prostate cancer diagnoses per 1000, as reported in the ERSPC study at 23 years, this corresponds to 4 fewer diagnoses per 1000 (95% CI 6 fewer to 1 fewer). We downgraded the certainty of the evidence for study limitations and inconsistency. We did not downgrade further for imprecision since we attributed the apparent imprecision to inconsistency. This interpretation was based on an MCID of 5 per 1000 men screened.

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Figure 9. PSA screening versus no screening: metastatic prostate cancer diagnosis



Heterogeneity: Tau² (REML[d]) = 0.03; Chi² = 4.58, df = 1 (P = 0.03); I² = 78%

Footnotes

[a]As randomised - 23 Year Follow-Up (core ages)
[b]As randomised - 16.9 Year Follow-Up
[c]CI calculated by Wald-type method.
[d]Tau² calculated by Restricted Maximum-Likelihood method.

Risk of bias legend

(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias): Subjective outcomes
(E) Incomplete outcome data (attrition bias): Metastatic prostate cancer diagnosis
(F) Selective reporting (reporting bias)
(G) Adherence
(H) Contamination
(I) Other bias

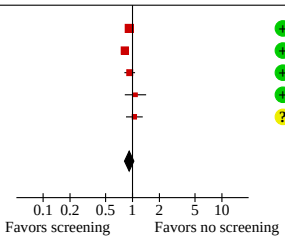
Sensitivity analysis: inclusion of the total ERSPC cohort

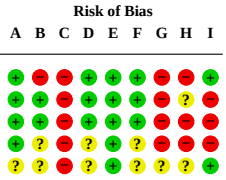
When we included the entire cohort of the ERSPC study, we found similar results for prostate cancer-specific mortality (Rate ratio 0.92, 95% CI 0.81 to 1.04; I^2 = 48%; 5 studies, 741,371 participants; Figure 10) and overall mortality (Rate ratio 0.99, 95% CI 0.97 to 1.00; I^2 = 0%; 4 studies, 694,989 participants; Figure 11). The certainty of evidence ratings were the same as that of the primary analysis (low) for these two outcomes. We were unable to analyse adverse events due to the lack of eligible trial data.

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Figure 10. PSA screening versus no screening: prostate cancer-specific mortality (ERSPC all ages)

Study or Subgroup	log[Rate Ratio]	SE	Screening Total	No screening Total	Weight	Rate Ratio IV, Random, 95% CI
CAP[a]	-0.083382	0.038896	189326	219395	32.9%	0.92 [0.85 , 0.99]
ERSPC[b]	-0.198	0.05	82816	99184	27.9%	0.82 [0.74 , 0.90]
PLCO[c]	-0.072	0.07	38340	38343	20.3%	0.93 [0.81 , 1.07]
Quebec[d]	0.076	0.14	31133	15353	7.7%	1.08 [0.82 , 1.42]
Stockholm[e]	0.048	0.11	2400	25081	11.2%	1.05 [0.85 , 1.30]
Total (HKSJ[f])			**344015**	**397356**	**100.0%**	**0.92 [0.81 , 1.04]**





Test for overall effect: T = 1.98, df = 4 (P = 0.12)

Heterogeneity: Tau² (REML[g]) = 0.00; Chi² = 7.57, df = 4 (P = 0.11); I² = 48%

Footnotes

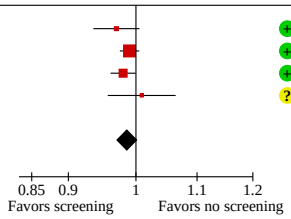
[a]As randomised - 15 Year Follow-Up
[b]As randomised - 16 Year Follow-Up (all ages)
[c]As randomised - 16.9 Year Follow-Up
[d]As randomised - 11 Year Follow-Up
[e]As randomised - 20 Year Follow-Up
[f]CI calculated by Hartung-Knapp-Sidik-Jonkman (HKSJ) method.
[g]Tau² calculated by Restricted Maximum-Likelihood method.

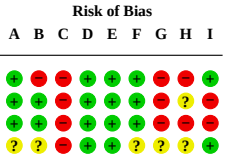
Risk of bias legend

(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias): Subjective outcomes
(E) Incomplete outcome data (attrition bias): Prostate cancer-specific mortality
(F) Selective reporting (reporting bias)
(G) Adherence
(H) Contamination
(I) Other bias

Figure 11. PSA screening versus no screening: overall mortality (ERSPC all ages)

Study or Subgroup	log[Rate Ratio]	SE	Screening Total	No screening Total	Weight	Rate Ratio IV, Random, 95% CI
CAP[a]	-0.030459	0.01832	189386	219439	9.5%	0.97 [0.94 , 1.01]
ERSPC[b]	-0.01	0.00769	82816	99184	54.1%	0.99 [0.98 , 1.01]
PLCO[c]	-0.02	0.01	38340	38343	32.0%	0.98 [0.96 , 1.00]
Stockholm[d]	0.009	0.027	2400	25081	4.4%	1.01 [0.96 , 1.06]
Total (HKSJ[e])			**312942**	**382047**	**100.0%**	**0.99 [0.97 , 1.00]**





Test for overall effect: T = 2.98, df = 3 (P = 0.06)

Heterogeneity: Tau² (REML[f]) = 0.00; Chi² = 2.16, df = 3 (P = 0.54); I² = 0%

Footnotes

[a]As randomised - 15 Year Follow-Up
[b]As randomised - 16 Year Follow-Up (all ages)
[c]As randomised - 16.9 Year Follow-Up
[d]As randomised - 20 Year Follow-Up
[e]CI calculated by Hartung-Knapp-Sidik-Jonkman (HKSJ) method.
[f]Tau² calculated by Restricted Maximum-Likelihood method.

Risk of bias legend

(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias): Objective outcomes
(E) Incomplete outcome data (attrition bias): Overall mortality
(F) Selective reporting (reporting bias)
(G) Adherence
(H) Contamination
(I) Other bias

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Sensitivity analysis: ERSPC sites as independent trials

We found similar results for prostate cancer-specific mortality in a pooled analysis that treated the ERSPC study sites as separate trials (ERSPC (Aargau) Switzerland; ERSPC (Antwerp) Belgium; ERSPC (Göteborg) Sweden; ERSPC (Rotterdam) Netherlands; ERSPC Finland; ERSPC France; ERSPC Italy; ERSPC Spain): Rate ratio 0.92, 95% CI 0.85 to 0.99; I^2 = 27%; 12 studies. The certainty of the evidence rating was the same (low) as that of the primary analysis for this outcome. Analyses for overall mortality and adverse events were not possible due to a lack of data.

Subgroup analyses: age

Based on the data stratified by age intervals reported in three trials (CAP; ERSPC; PLCO), we compared prostate cancer-specific mortality in men younger than 65 years versus those 65 years and older. Effect sizes were similar, with low heterogeneity (I^2 = 0%), and the test for subgroup differences was not statistically significant (P = 0.66). There were insufficient data to conduct subgroup analyses for overall mortality and adverse events.

Subgroup analyses: screening interval

Based on the available evidence from five trials, we compared prostate cancer-specific mortality between annual screening (PLCO; Quebec), biennial screening (ERSPC), and one-time screening (CAP; Stockholm). There was considerable heterogeneity across the studies and the test for subgroup differences was statistically significant (P = 0.04), with lower prostate cancer-specific mortality in the biennial group (Analysis 1.12). For overall mortality, the test for subgroup differences was not statistically significant (P = 0.78) (Analysis 1.13). There were insufficient data to conduct subgroup analyses for adverse events.

Comparison 2. Screening based on PSA (alone) – sensitivity analysis based on risk of bias

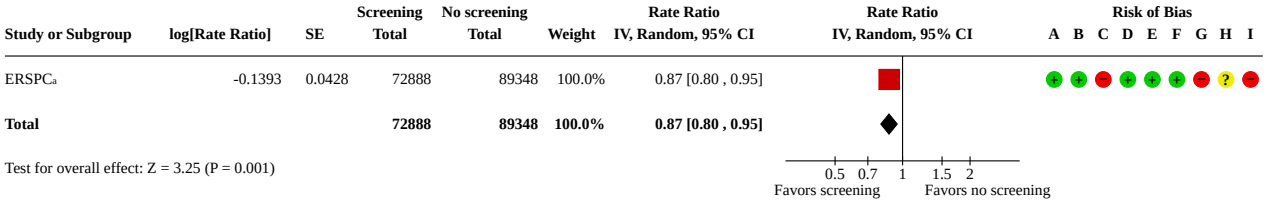
We conducted predefined subgroup analyses based on the ERSPC study, which stood out for a lower risk of bias than the other included trials. We report the results for this single trial in Summary of findings 2.

Critical outcomes

Prostate cancer-specific mortality

Screening likely reduces prostate cancer-specific mortality (Rate ratio 0.87, 95% CI 0.80 to 0.95; 1 study, 162,236 participants; moderate-certainty evidence; Figure 12). Assuming a baseline risk of 16 prostate cancer-related deaths per 1000, as reported by the ERSPC study at 23 years, this corresponds to 2 fewer deaths (95% CI 3 fewer to 1 fewer) per 1000. We downgraded the certainty of the evidence due to study limitations. This interpretation was based on an MCID of 1 per 1000 men screened.

Figure 12. PSA screening versus no screening - sensitivity analysis based on risk of bias: prostate cancer-specific mortality



Test for overall effect: Z = 3.25 (P = 0.001)

Heterogeneity: Not applicable

Footnotes

aAs randomised - 23 Year Follow-Up (core ages)

Risk of bias legend

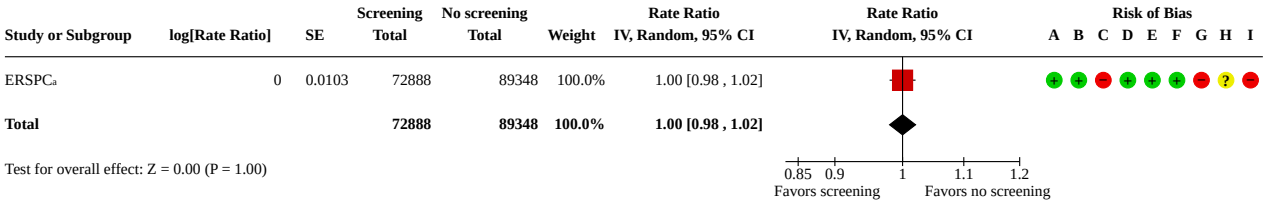
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias): Subjective outcomes
(E) Incomplete outcome data (attrition bias): Prostate cancer-specific mortality
(F) Selective reporting (reporting bias)
(G) Adherence
(H) Contamination
(I) Other bias

Overall mortality

We are very uncertain as to how screening impacts overall mortality based on this trial (Rate ratio 1.00, 95% CI 0.98 to 1.02; 1 study, 162,236 participants; very low-certainty evidence; Figure 13). We downgraded the certainty of the evidence due to serious study limitations and very serious imprecision (the 95% confidence interval was consistent with both no effect and increased overall mortality). This interpretation was based on an MCID of 1 per 1000 men screened.

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Figure 13. PSA screening versus no screening - sensitivity analysis based on risk of bias: overall mortality



Test for overall effect: Z = 0.00 (P = 1.00)

Heterogeneity: Not applicable

Footnotes

aAs randomised - 23 Year Follow-Up (core ages)

Risk of bias legend

(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias): Objective outcomes
(E) Incomplete outcome data (attrition bias): Overall mortality
(F) Selective reporting (reporting bias)
(G) Adherence
(H) Contamination
(I) Other bias

Adverse events

Not reported.

Important outcomes

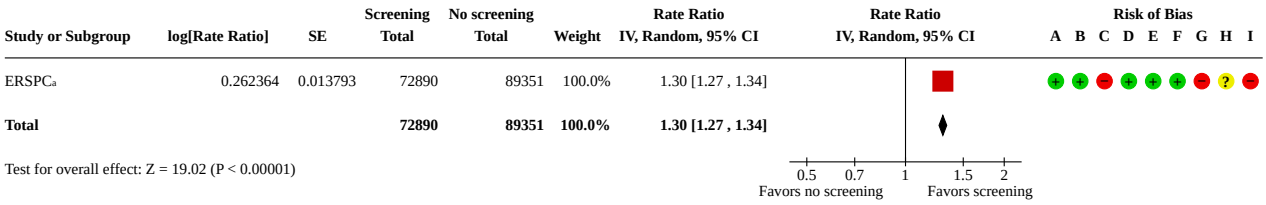
Quality of life

See the data reported above on the ERSPC Finland substudy.

Prostate cancer diagnosis

Screening likely increases prostate cancer diagnosis (Rate ratio 1.30, 95% CI 1.27 to 1.34; 1 study, 162,241 participants; moderate-certainty evidence; Figure 14). Assuming a baseline risk of 120 prostate cancer diagnoses per 1000, as reported by the ERSPC study at 23 years, this corresponds to 36 more diagnoses (95% CI 32 more to 41 more) per 1000. This interpretation was based on an MCID of 10 per 1000 men screened.

Figure 14. PSA screening versus no screening - sensitivity analysis based on risk of bias: prostate cancer diagnosis



Test for overall effect: Z = 19.02 (P < 0.00001)

Heterogeneity: Not applicable

Footnotes

aAs randomised - 23 Year Follow-Up (core ages)

Risk of bias legend

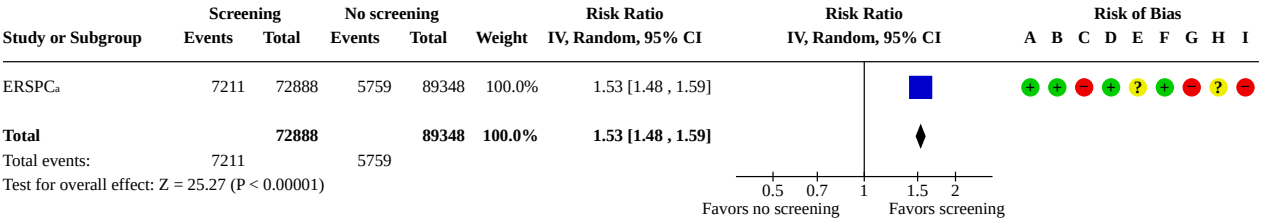
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias): Objective outcomes
(E) Incomplete outcome data (attrition bias): Prostate cancer diagnosis
(F) Selective reporting (reporting bias)
(G) Adherence
(H) Contamination
(I) Other bias

Localised (stage I and II) prostate cancer diagnosis

Screening likely increases localised (stage I and II) prostate cancer diagnosis (RR 1.53, 95% CI 1.48 to 1.59; 1 study, 162,236 participants; moderate-certainty evidence; Figure 15). Assuming a baseline risk of 64 prostate cancer diagnoses per 1000, as reported by the ERSPC study at 23 years, this corresponds to 34 more localised prostate cancer diagnoses (95% CI 31 more to 38 more) per 1000. This interpretation was based on an MCID of 10 per 1000 men screened.

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Figure 15. PSA screening versus no screening - sensitivity analysis based on risk of bias: localised (stage I and II) prostate cancer diagnosis



Heterogeneity: Not applicable

Footnotes

[a]As randomised - 23 Year Follow-Up (core ages) - low and intermediate risk categories.

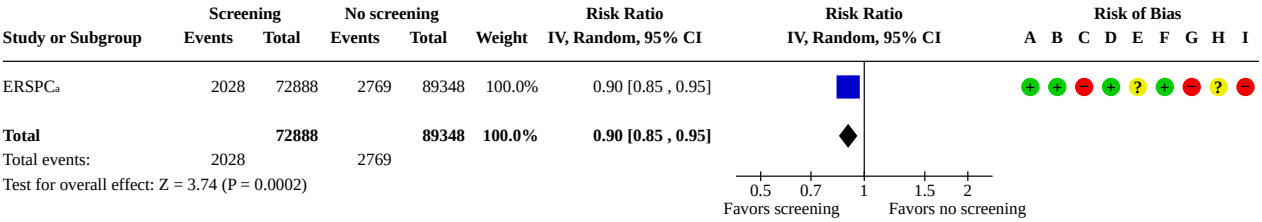
Risk of bias legend

(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias): Subjective outcomes
(E) Incomplete outcome data (attrition bias): Localized (stage I and stage II) prostate cancer diagnosis
(F) Selective reporting (reporting bias)
(G) Adherence
(H) Contamination
(I) Other bias

Advanced (stage III and IV) prostate cancer diagnosis

Screening likely reduces advanced (stage III and IV) prostate cancer diagnosis (RR 0.90, 95% CI 0.85 to 0.95; 1 study, 162,236 participants; moderate-certainty evidence; Figure 16). Assuming a baseline risk of 31 advanced prostate cancer diagnoses per 1000, as reported by the ERSPC study at 23 years, this corresponds to 3 fewer advanced prostate cancer diagnoses (95% CI 5 fewer to 2 fewer) per 1000. This interpretation was based on an MCID of 10 per 1000 men screened.

Figure 16. PSA screening versus no screening - sensitivity analysis based on risk of bias: advanced (stage III and IV) prostate cancer diagnosis



Heterogeneity: Not applicable

Footnotes

[a]As randomised - 23 Year Follow-Up (core ages) - high-risk category.

Risk of bias legend

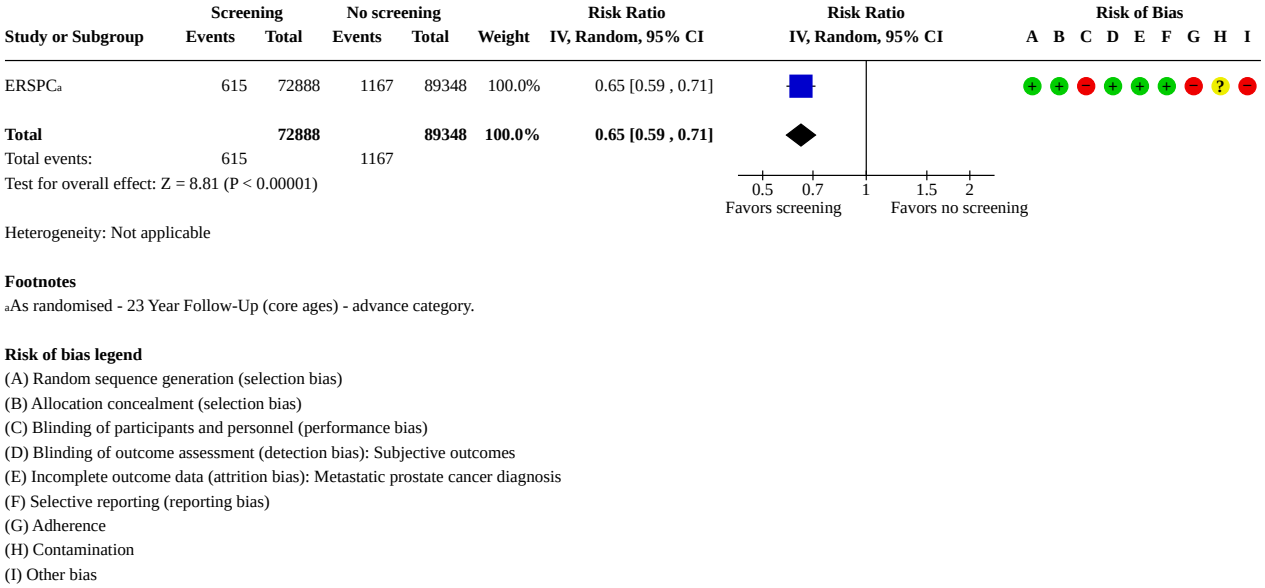
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias): Subjective outcomes
(E) Incomplete outcome data (attrition bias): Advanced (stage III and IV) prostate cancer diagnosis
(F) Selective reporting (reporting bias)
(G) Adherence
(H) Contamination
(I) Other bias

Metastatic prostate cancer diagnosis

Screening may reduce the diagnosis of metastatic prostate cancer disease (RR 0.65, 95% CI 0.59 to 0.71; 1 study, 162,236 participants; low-certainty evidence; Figure 17). Assuming a baseline risk of 13 diagnoses of metastatic prostate cancer, as reported by the ERSPC study at 23 years, this corresponds to 5 fewer metastatic prostate cancer diagnoses (95% CI 5 fewer to 4 fewer) per 1000. This interpretation was based on an MCID of 5 per 1000 men screened. We downgraded the certainty of the evidence due to study limitations.

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Figure 17. PSA screening versus no screening - sensitivity analysis based on risk of bias: metastatic prostate cancer diagnosis



Footnotes

[a]As randomised - 23 Year Follow-Up (core ages) - advance category.

Risk of bias legend

(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias): Subjective outcomes
(E) Incomplete outcome data (attrition bias): Metastatic prostate cancer diagnosis
(F) Selective reporting (reporting bias)
(G) Adherence
(H) Contamination
(I) Other bias

Comparison 3. Screening based on PSA, kallikrein panel, and MRI versus control

See Summary of findings 3. We did not include this study in the main comparison above because it combines PSA with other downstream tests, which were not considered in the other trials.

Critical outcomes

Prostate cancer-specific mortality, overall mortality, and adverse events

Not reported.

Important outcomes

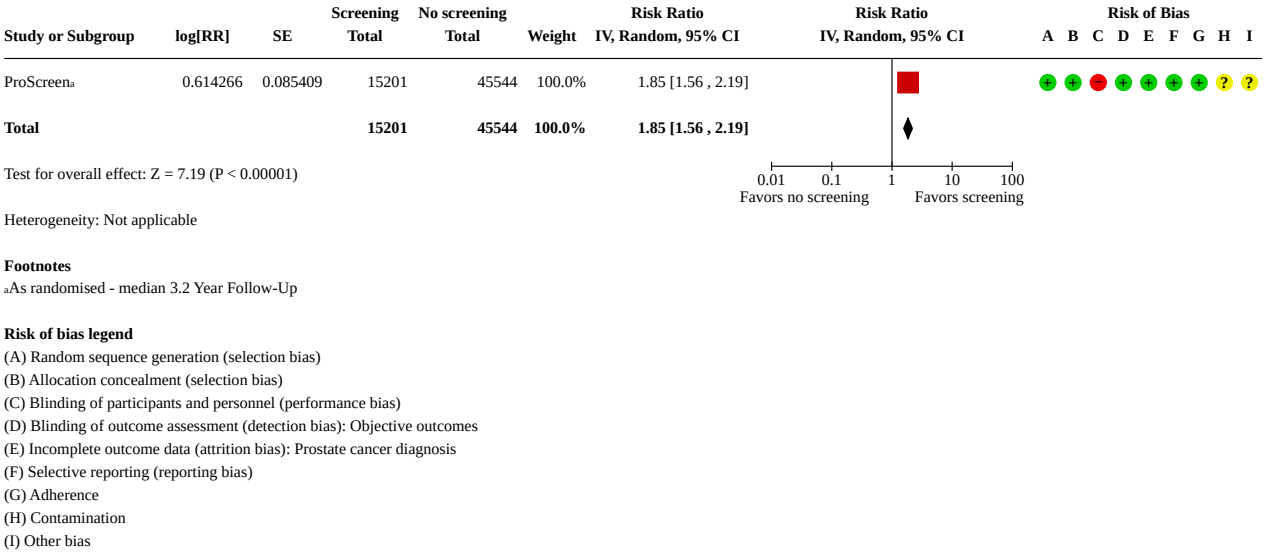
The ProScreenstudy did not report data for quality of life or metastatic prostate cancer diagnosis.

Prostate cancer diagnosis

Screening based on PSA, kallikrein panel, and MRI likely has little to no effect on prostate cancer diagnosis compared to control (RR 1.85, 95% CI 1.56 to 2.19; 1 study, 60,745 participants; moderate-certainty evidence; Figure 18). Assuming a baseline risk of 8 prostate cancer diagnoses per 1000, this corresponds to a risk difference of 6 more diagnoses (95% CI 4 more to 9 more) per 1000 individuals after a median follow-up of 3.2 years. We downgraded the certainty of the evidence for study limitations. This interpretation was based on an MCID of 10 per 1000 men screened.

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Figure 18. PSA/kallikrein/MRI-based screening versus no screening: prostate cancer diagnosis



Footnotes

[a]As randomised - median 3.2 Year Follow-Up

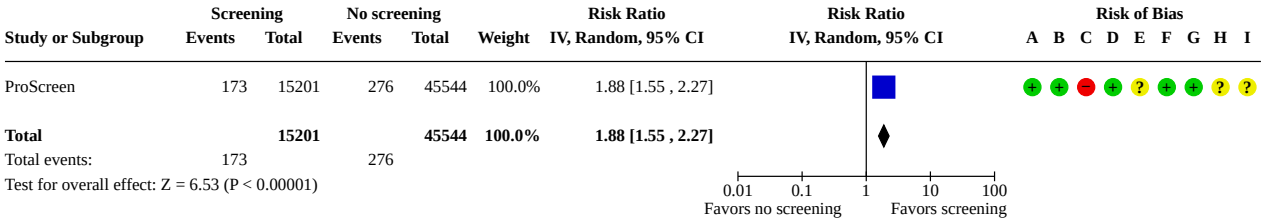
Risk of bias legend

(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias): Objective outcomes
(E) Incomplete outcome data (attrition bias): Prostate cancer diagnosis
(F) Selective reporting (reporting bias)
(G) Adherence
(H) Contamination
(I) Other bias

Localised prostate cancer diagnosis

Screening based on PSA, kallikrein panel, and MRI likely has little to no effect on localised prostate cancer diagnoses compared to control (RR 1.88, 95% CI 1.55 to 2.27; 1 study, 60,745 participants; moderate-certainty evidence; Figure 19). Assuming a baseline risk of 6 localised prostate cancer diagnoses per 1000, this corresponds to a risk difference of 5 more diagnoses (95% CI 3 more to 8 more) per 1000 after a median follow-up of 3.2 years. We downgraded the certainty of the evidence for study limitations. This interpretation was based on an MCID of 10 per 1000 men screened.

Figure 19. PSA/kallikrein/MRI-based screening versus no screening: localised prostate cancer diagnosis (T1-T2 N0/Nx M0/Mx)



Risk of bias legend

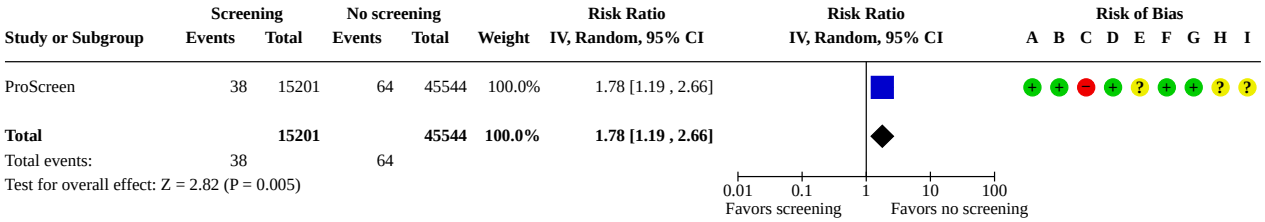
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias): Subjective outcomes
(E) Incomplete outcome data (attrition bias): Localized (stage I and stage II) prostate cancer diagnosis
(F) Selective reporting (reporting bias)
(G) Adherence
(H) Contamination
(I) Other bias

Advanced prostate cancer diagnosis

Screening based on PSA, kallikrein panel, and MRI likely has little to no effect on advanced prostate cancer diagnosis compared to control (RR 1.78, 95% CI 1.19 to 2.66; 1 study, 60,745 participants; moderate-certainty evidence; Figure 20). Assuming a baseline risk of 1 advanced prostate cancer diagnosis per 1000, this corresponds to a risk difference of 2 more diagnoses (95% CI 1 more to 2 more) per 1000 after a median follow-up of 3.2 years. We downgraded the certainty of the evidence for study limitations. This interpretation was based on an MCID of 10 per 1000 men screened.

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Figure 20. PSA/kallikrein/MRI-based screening versus no screening: advanced prostate cancer diagnosis (T3–T4 N1 M0/MX, TX NX M1)



Risk of bias legend
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias): Subjective outcomes
(E) Incomplete outcome data (attrition bias): Advanced (stage III and IV) prostate cancer diagnosis
(F) Selective reporting (reporting bias)
(G) Adherence
(H) Contamination
(I) Other bias

Reporting biases

We were unable to produce funnel plots due to the few studies included in each analysis. We consider it unlikely that we missed results as trials in this area are large-scale and usually yield multiple publications covering both primary and secondary analyses.

DISCUSSION

Summary of main results

This review's findings are based on 721,612 participants from five studies comparing prostate-specific antigen (PSA)-based prostate cancer screening with no screening and 60,745 participants from one study comparing a combination of PSA, kallikrein panel, and magnetic resonance imaging (MRI)-based screening with no screening.

PSA-based screening versus no screening

Based on evidence of moderate to low certainty, PSA-based prostate cancer screening may reduce both prostate cancer-specific mortality and overall mortality. However, it may increase the number of men diagnosed with prostate cancer overall, and those with localised disease (stage I and II). In contrast, the incidence of advanced prostate cancer (stages III and IV) and metastatic prostate cancer diagnosis may be similar between the screened and unscreened participants. Adverse events – defined as deaths related to prostate biopsy or prostate cancer treatment – also appear to be similar between screening and no screening. There may be little to no effect of screening on men's quality of life.

A predefined sensitivity analysis of the ERSPC trial, which was assessed as having the lowest risk of bias, found that PSA-based screening probably reduces prostate cancer-specific mortality. In this trial, the effect on overall mortality was very uncertain. Prostate cancer diagnoses and localised prostate cancer diagnoses are likely to increase, advanced prostate cancer diagnoses are likely reduced, and metastatic prostate cancer diagnoses may be reduced. As above, there may be little to no effect of screening on men's quality of life.

PSA, kallikrein panel, and MRI-based screening versus no screening

Based on moderate-certainty evidence from a single study with a median follow-up of 3.2 years (ProScreen), screening using PSA, kallikrein panel, and MRI probably has little to no effect on the overall detection of prostate cancer, including both localised (T1–T2N0/Nx M0/Mx) and advanced (T3–T4N1 M0/Mx, TX NX M1) stages, compared to no screening.

Limitations of the evidence included in the review

The certainty of our effect size estimates by outcome ranged from moderate to very low. The main issues that lowered our confidence in the pooled screening effect were study limitations, such as lack of allocation concealment, performance bias, and attrition bias. Other issues that lowered our confidence in the estimates of the effect of individual outcomes were mainly inconsistency and imprecision.

Nevertheless, this Cochrane review summarises a formidable body of evidence summarising six randomised trials with over 700,000 participants. It includes all relevant randomised controlled trials (RCTs) on prostate cancer screening but excludes the Norrkoping study [35] (included in prior versions of this review) due to its pseudo-randomised design (every other participant was assigned to screening). Given the considerable challenges in interpreting observational studies in PSA-based prostate screening [294], we did not include non-randomised trials in accordance with our original protocol [51]. We did not find any RCTs that evaluated the role of digital rectal examination (DRE) alone in screening for prostate cancer. However, many of the included studies evaluated DRE in combination with PSA screening.

The following issues deserve consideration.

- The effects of prostate cancer screening using magnetic resonance imaging [18, 295]. A recent Cochrane review demonstrated the potential benefits of an integrated MRI pathway [14]. However, no trials of PSA-based screening integrating MRI are available yet. Two trials are ongoing but are not expected to provide results for nearly 10 years. In addition,

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there is major research interest in genetic markers, but their role is far less well established.

- Existing prostate cancer screening trials have been nearly exclusively conducted in White men. Black men appear to be at higher risk of developing prostate cancer and may also have a worse prognosis. The effect and utility of prostate cancer screening may be different in Black men. However, very few non-White men have been enrolled in these screening trials. Of note, a large retrospective cohort study indicated that when Black and White men with prostate cancer receive comparable care, outcomes are similar, suggesting that inequities such as access to care, rather than inherent biology, drive observed differences at the population level [296]. Similarly, it is unclear what effect prostate cancer screening has in men with a positive family history of prostate cancer or prior exposure to Agent Orange [297], both of which are associated with increased risk [298], as these risk factors were not captured in existing trials.
- The time horizon of this review is limited by the follow-up duration of included trials. We did not model or extrapolate outcomes beyond this period (as others have) [299], as doing so would introduce additional uncertainty. Extended follow-up of existing trials will provide the best evidence on the long-term impact of PSA screening. This is especially true for ProScreen, for which only very early results are available. Study completion and final results are not expected until 2038.
- Data on the outcomes of advanced prostate cancer and metastatic prostate cancer in this review are limited to the stage of diagnosis (not the cumulative incidence that should include disease progression after diagnosis and initial management). Since screening can advance diagnoses without affecting treatment outcomes, these may be biased outcome measures [300]. Disease-specific mortality and overall mortality are therefore the most robust outcome measures of prostate cancer screening trials.
- In contrast to other systematic reviews – in particular, one by Fenton and colleagues [44] – the reported analysis is limited to evidence from RCTs and direct evidence for its downstream effects. Existing studies do not provide information about treatment-related complications such as urinary incontinence, erectile dysfunction, bowel-related issues, and quality of life, and it may be argued that these issues exceed what can be expected to be reported in a PSA screening trial. Such information is available from several pivotal treatment trials, such as the SPCG-4, PIVOT, and ProtecT studies [301, 302, 303], but its integration would require at least some basic modelling that was outside the scope of this review. It is therefore a limitation of this review that we did not systematically review and include the interventions that would ultimately be responsible for producing downstream changes in all-cause and disease-specific mortality.
- In contrast to other published reviews, but in accordance with prior versions of this review [41, 304], we did not seek to quantify the degree of overdiagnosis; this is a limitation of this review. The most recent estimate found a rate of 35 (95% CI 13 to 56) to 60 (95% CI 54 to 66) per 1000 invited men when using a PSA cut-off of below 4 ng/mL [305]. Overdiagnosis adds to the harm and burden aspect of screening.
- All included trials in this systematic review included various degrees of contamination in the control arm (see Figure 2). This issue was most prominent in the PLCO trial, which has thus been described as a trial comparing high-intensity to low-intensity screening [243]. The actual comparator in this trial therefore included varying degrees of PSA testing but no organised screening.

Limitations of the review processes

We conducted this review in accordance with the current Cochrane standards for high-quality systematic reviews. However, there is still the potential for bias, as follows.

- The original protocol for this review was published over 20 years ago and is considerably less detailed than a more contemporary protocol would be [51], specifically with regard to secondary analyses (subgroup and sensitivity analyses). Overall, we believe that this updated review and its prior versions have used a consistent approach that minimises bias.
- Our review continues to use the original Cochrane risk of bias tool [306], rather than the updated and methodologically advanced Cochrane risk of bias 2 tool [307]. We choose to do so to maintain continuity with prior versions of the review [33, 75]. We assessed additional domains (adherence and contamination), which are of major importance in this context. We believe that the resulting risk of bias judgments and GRADE certainty of evidence ratings which draw on them are appropriate and correct.
- A critical aspect in the interpretation of this body of evidence was the thresholds chosen for MCIDs, which are essential to the application of GRADE. We established these MCIDs prior to the data analyses, but they were not captured in our protocol [51]. We did not predefine thresholds to categorise effect sizes as 'trivial', 'small', 'moderate', or 'large', which may have limited the usefulness of this review for some users and guideline developers [308, 309]. However, we provide transparent reporting of all judgements, allowing readers to apply their own thresholds if desired. Where thresholds were used, these were informed by the content experts involved in the review (PD, JF, ECH, and JHJ), rather than derived from emerging methodological approaches [310], and are consistent with those used in another systematic review by the same research team and in a guideline [75, 311].
- A controversial aspect of meta-analysis is the degree of allowable heterogeneity across pooled trials. Unlike other systematic review authors [44], we see value in providing pooled analyses of trials that met predefined inclusion criteria, thereby allowing readers to assess how results compared in the different trials.
- We used the latest models recommended in the *Cochrane Handbook for Systematic Reviews of Interventions* to perform a random-effects meta-analysis, which, in some cases, resulted in estimates with wide confidence intervals [78]. This is a limitation of these models, which provide conservative estimates but may result in overly wide confidence intervals in the presence of high statistical heterogeneity and few studies (Analysis 1.4; Analysis 1.5).
- We considered the ERSPC study to have an overall 'lower' risk of bias than the other included studies, while acknowledging that we judged it to have a high risk in some domains. This assessment might underplay the study's limitations insofar as the evidence for some outcomes had a higher certainty rating in the sensitivity analyses. Nonetheless, for transparency, we presented the estimates for the whole body of evidence and the trial separately, in case policymakers wish to use the

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estimates from the pooled data rather than those derived from the sensitivity analyses.

- Screening for prostate cancer has no known beneficial influence on causes of death other than prostate cancer; therefore, the benefits seen on overall mortality should not exceed those seen for disease-specific mortality. The apparent differences are likely related to the manner in which absolute effect size estimates are calculated by multiplying a relative effect size with a baseline risk. The true effect size likely corresponds to the absolute effect size estimate for disease-specific mortality, which was labelled as 'moderate certainty of evidence'.
- We included randomised controlled trials that primarily aimed to assess the effectiveness of screening, with limited data on adverse events which can be captured by non-randomised studies looking at the downstream testing and treatment of participants following screening.
- While we deemed the GÖTEBORG-2 2022 trial to be eligible for inclusion (and have listed it as an ongoing study in Supplementary material 4), we excluded two other important trials: STHLM3-MRI [312, 313], which focused on the diagnostic accuracy of alternative testing strategies [314], and PROBASE 2013 [315, 316, 317, 318], which did not include a non-PSA screening arm [319].

Agreements and disagreements with other studies or reviews

Here we outline the key differences between our findings and the most relevant high-quality systematic reviews on prostate cancer screening.

This review was last updated in 2013 [33], before the publication of the CAP trial, and was based on short-term outcomes of the included trials, in particular the ERSPC trial. In addition to the inclusion of the CAP trial, we also added (preliminary) results of the ongoing ProScreen trial. The 2013 version of this review included the (small) Norkopping trial [35], which we excluded from this update for methodological reasons. The 2013 version of the review focused on the overall body of evidence, but mentioned the diverging results of the ERSPC. It concluded that PSA screening conferred "no significant reduction in prostate cancer-specific and overall mortality" [33], which differs substantially from our current findings.

Our research team authored a systematic review on prostate cancer screening published in *The BMJ* in 2018 [75], resulting in some overlap in methodology between the *BMJ* review and this one. The *BMJ* review informed a BMJ Rapid Recommendation for clinical practice [311], which lies outside the scope of this present review. This Cochrane review update extends the *BMJ* review in several ways. These include conducting a secondary analysis using all available participants from the ERSPC trial; performing an additional analysis in which we treated each individual ERSPC trial site as a separate study; incorporating extended 23-year follow-up data from the ERSPC trial; and including data from the ProScreen trial.

Rahal and colleagues published a 'living' systematic review in 2016 [320], but the review has not been updated since then. Overall, they found a significant benefit for screening, with a risk ratio of 0.78 (95% CI 0.65 to 0.94; I^2 = 0%, 95% CI 0% to 69%) and a number needed to invite for screening (to prevent one prostate cancer death) of 1000 men. The authors did not rate the certainty of the evidence. They concluded that screening of adequate duration reduces mortality from prostate cancer and that the benefit, while small, compared favourably with screening for other cancers.

The most notable recent systematic review is by Fenton and colleagues [44]. It informed the most recent version of the US Preventive Services Task Force recommendations on prostate cancer screening [321]. The Fenton review did not include the ERSPC's recently published long-term follow-up data or the CAP trial, as these were published after the date of their search. It excluded the Quebec and Stockholm studies because of low methodological quality. While we appreciate these concerns, we included these trials in order to synthesise the totality of available evidence. When assessing the efficacy of PSA screening for prostate cancer-specific mortality and overall mortality, the Fenton meta-analysis is based on only two trials (ERSPC; PLCO). To assess the adverse effects of screening (such as biopsy-related complications and downstream harm of treatment), for which there is little direct evidence from the available trials, the authors' analysis was based on observational studies, such as SPCG-4 [301], PIVOT [303], and ProtecT [302], which have assessed the comparative effectiveness of different prostate cancer treatment modalities. We excluded these studies because our systematic review focused on randomised controlled trials only. The Fenton review also sought to establish the harms of prostate cancer screening by investigating the adverse consequences of major treatment trials such as SPCG-4 [301], PIVOT [303], and ProtecT [302], as well as observational studies. In summary, the authors concluded that "active treatments for screen-detected prostate cancer have unclear effects on long-term survival but are associated with sexual and urinary difficulties."

The German Institute for Quality and Efficiency in Healthcare issued an evidence report on PSA-based prostate cancer screening and its assessment of a potential net benefit [305]. It shares with our review a focus on randomised controlled trials only. It differs in its stricter exclusion criteria that led to the exclusion of several important studies, such as CAP. It did not rate the certainty of the evidence and used statistical (rather than clinical) significance as the framework for interpretation. For all-cause mortality, there was no "indication" of either a benefit or harm. For disease-specific mortality and the development of metastases, there was an indication of a benefit. There was also no evidence that screening affects quality of life, adverse effects, or the consequences of false-positive screening results. However, this was due to the lack of eligible RCT data. Finally, the report found "proof" of harm due to large numbers of false-positive PSA tests leading to unnecessary biopsies, and considerable overdiagnosis leading to unnecessary treatment. Ultimately, the review found that the potential benefits of PSA-based prostate cancer screening did not outweigh its potential harms.

The United Kingdom National Screening Committee (UK NSC) recently revisited its current recommendation against population-based screening in the UK and initiated an external review and reappraisal of the evidence. This report is now in its final version [322]. For the assessment of the impact of PSA screening on prostate cancer morbidity, mortality, and overall mortality, the three most recent major trials were included (CAP; ERSPC; PLCO). The discussion emphasises the inconsistent findings of these trials and the major issue of control arm contamination. When

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evaluating whether screening using risk algorithms or the inclusion of markers other than PSA alone can better identify men with clinically significant prostate cancer, evidence from this report suggested that MRI (either added to PSA-based screening or as a standalone modality) and the Stockholm 3 predictive model may offer improved diagnostic accuracy [323]. Regarding the potential harms and benefits of currently available treatment approaches for early-stage prostate cancer, the review identified a lack of evidence distinguishing the effects of treatment for low-, intermediate-, and high-risk disease with direct comparisons. The investigators concluded that it therefore remains unclear "whether the potential benefits of radical treatments on disease progression in comparison to observation can offset the increased rate of adverse events, particularly for men who may never have clinically important disease." In summary, the report states that PSA-based screening of asymptomatic men is still not recommended.

AUTHORS' CONCLUSIONS

Implications for practice

Based on this updated review and meta-analyses, which yielded mostly low- to moderate-certainty evidence across outcomes, prostate-specific antigen (PSA)-based screening likely reduces prostate cancer-specific mortality and overall mortality. However, the absolute benefit is small. At the same time, the number of men diagnosed with prostate cancer has markedly increased, potentially exposing large numbers of men to overdiagnosis and overtreatment. These potential adverse downstream effects of PSA-based screening are incompletely represented in this review (since the outcomes are not captured in the included trials) and would require informal or formal modelling to quantify, as others have done [44, 299, 311]. Emerging screening alternatives, such as with a kallikrein panel and magnetic resonance imaging (MRI), may have little to no effect on diagnoses of prostate cancer, but the results on mortality are not yet known.

Assessing the trade-offs between the potential benefits and harms of PSA screening, including patient values and preferences, resource use, and issues of acceptability, feasibility, and equity, is beyond the scope of this review. Such judgements are the domain of evidence-based guideline developers, such as the United States Preventive Services Task Force, which is currently updating its recommendations [324]. Relevant considerations will include the following.

- The extent to which benefits for a relatively small number of individuals (per 1000 asymptomatic men screened) outweigh the burden, harms, and costs to a much larger group of individuals [325]. Also, how is this balance impacted by the continuously broadening treatment options for men with advanced prostate cancer resulting in improved outcomes [25, 326, 327, 328].
- The increasing integration of MRI in the screening pathway. Originally developed specifically to aid in the triage process of men with an elevated PSA and to reduce the risk of overdiagnosis [14, 18], it is also often misused to maximise the diagnostic yield [329], rather than reducing the diagnosis of low-risk prostate cancer.
- The fact that uptake of active surveillance for low-risk prostate cancer has risen from approximately 25% in 2014 to 60% in 2021 in the USA [330]. In Sweden, the reported use of active surveillance increased from 2009 to 2014 from 57% to 91% for very low-risk prostate cancer and from 40% to 74% for low-risk prostate cancer [331]. Meanwhile, active surveillance use appears much lower in Germany, with one study reporting that only 7.7% of men under the age of 60 years choose this approach [332]. Rates of active surveillance are relevant to guideline developers and policymakers because of the downstream treatment burden on men who have been potentially overdiagnosed with prostate cancer.
- Advances in the biopsy procedure with a shift away from transrectal to transperineal biopsy, which may reduce infection rates even if antibiotics are omitted [333]; this is the subject of a separate, forthcoming Cochrane review [334]. Also, although there is a lack of evidence from randomised trials on the topic [335], the widespread adoption of robotic-assisted radical prostatectomy by a new generation of well-trained surgeons likely contributes to a lower treatment burden than in the past [336].

One issue that appears uncontroversial today, as reflected in all major guidelines [46, 47, 324, 337, 338], is the need for shared decision-making in asymptomatic men who are considering the PSA screening test. Ensuring that shared decision-making does take place, ideally with the support of a decision aid [339, 340, 341], is a critically important implementation consideration in any new or updated guideline recommendation.

Implications for research

Decision-making on PSA screening should be grounded in high-quality randomised controlled trials, which are complex and resource-intensive. Ongoing studies investigating advances – such as kallikrein-based testing and MRI in the ongoing ProScreen trial [43] – are therefore of paramount importance and should receive strong support from professional organisations, patient advocacy groups, and funding organisations. We also support the continued active follow-up and data accrual of pivotal trials in this field, including ERSPC [110], CAP [CAP], PIVOT [342], and ProTect [343].

A longstanding issue of debate has been whether screening guidelines for Black men should differ from those for White men, given evidence suggesting a higher incidence of prostate cancer and a potentially increased risk of more aggressive disease in the former group [4, 5]. Future trials should seek to include representative numbers of diverse patient populations to help inform their risk and management. In particular, future screening trials should seek to enrol representative numbers of Black men to better inform screening recommendations.

There is also considerable promise in genetic testing – for example, through the development of a polygenic risk score [9]. The benefits and risks of such new strategies deserve to be studied in rigorous, large, population-based trials before finding entry into routine care.

SUPPLEMENTARY MATERIALS

Supplementary materials are available with the online version of this article: 10.1002/14651858.CD004720.pub4.

Supplementary material 1 Search strategies

Supplementary material 2 Characteristics of included studies

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Supplementary material 3 Characteristics of excluded studies

Supplementary material 4 Characteristics of ongoing studies

Supplementary material 5 Analyses

Supplementary material 6 Data package

Supplementary material 7 Survey of trial investigators providing information on included trials

ADDITIONAL INFORMATION

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Editorial and peer-reviewer contributions

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The following people conducted the editorial process for this article:

- Sign-off Editor (final editorial decision): Toby Lasserson, Cochrane Evidence Production and Methods Directorate;
- Managing Editor (selected peer reviewers, provided editorial guidance to authors, edited the article): Hannah Payne, Cochrane Central Editorial Service;
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Contributions of authors

All review authors read and approved the final review draft.

ECH: reviewed abstracts and full-text studies for inclusion, performed data extraction and analysis, and co-led the write-up of this update.

JVAF: reviewed abstracts and full-text studies for inclusion, performed data extraction and analysis, and co-led the write-up of this update.

DI: initiated the review and wrote the initial protocol. He conducted the original literature search, reviewed abstracts and full-text studies for inclusion, performed quality assessment, data extraction, analysis, and contributed to the writing of the review.

JHJ: reviewed abstracts and full-text studies for inclusion, performed data extraction and analysis, and contributed to the writing of the review.

OV: performed data extraction, contributed to setting up the analysis and summary of findings tables, and wrote sections of the review.

AC: updated the current search strategy and performed the literature review; contributed to the writing of the review.

PD: co-initiated and oversaw the review update, reviewed abstracts and full-text studies for inclusion, performed quality assessment and data analysis, and contributed to the writing of the review.

Some authors involved in previous published versions of this review in 2013 are no longer included in the author byline: Molly M Neuberger, Mia Djulbegovic. Some of the content retained in this review reflects their contributions.

Declarations of interest

JVAF: Editor for Cochrane Urology and Senior Editor of the Cochrane Database of Systematic Reviews but was excluded from the editorial processing of this review.

ECH: Editor for Cochrane Urology, but was excluded from the editorial processing of this review.

DI: Editor for Cochrane Urology, but was excluded from the editorial processing of this review.

JHJ: Editor for Cochrane Urology, but was excluded from the editorial processing of this review.

OV: None.

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PD: Co-ordinating Editor of Cochrane Urology; however, he was not involved in the editorial process of this review.

ECH, JVAF, JHJ, and PD are healthcare professionals.

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Data, code and other materials

As part of the published Cochrane review, the following is made available for download for users of the Cochrane Library: full search strategies for each database (Supplementary material 1); full citations of each unique report for all studies included (Supplementary material 2), and ongoing and excluded studies at the full-text screen in the final review (Supplementary material 3; Supplementary material 4); study data, including study information, study arms, and study results or test data; consensus risk of bias assessments; and analysis data, including overall estimates and settings, subgroup estimates, and individual data rows (Supplementary material 5; Supplementary material 6), and survey administered to study authors (Supplementary material 7). Appropriate permissions have been obtained for such use. Analyses and data management were conducted using Cochrane's authoring tool, RevMan, which utilises built-in computational methods.

What's new

Date	Event	Description
15 May 2026	New citation required and conclusions have changed	Evidence from long-term follow-up of the main trials was added, indicating a change in the conclusions for the critical outcomes of the review.
15 May 2026	New search has been performed	Search updated; new studies and extended follow-up data added.

History

Protocol first published: Issue 2, 2004
Review first published: Issue 3, 2006

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ADDITIONAL TABLES

Table 1. Overview of studies included in synthesis (study characteristics, intervention, and comparisons)

Study name	Trial period (year to year)	Setting/country	Description of participants	Intervention(s) and comparator(s)	Duration of intervention (duration of follow-up)	Age (median years)	Outcomes in the summary of findings tables
CAP	2002-2021	United Kingdom	All men aged 50 to 69 years in each of the randomised primary care practices	Screening	15 years	58.5	PCa mortality, overall mortality, adverse events, PCa diagnosis, localised and advanced PCa diagnosis
				No screening		58.6	
ERSPC	1993-	Multicentre across: Belgium, Finland, France, Italy, the Netherlands, Spain, Sweden, Switzerland (Portugal withdrew from the study in October 2000)	Men aged 50 to 74 years from participating sites	Screening	23 years	60.2 (core group)	PCa mortality, overall mortality, PCa diagnosis, localised and advanced PCa diagnosis, metastatic PCa diagnosis
				No screening		60.2 (core group)	
ERSPC (Antwerp) Belgium	1992-	Belgium	Men aged 55 to 74 years	Screening	13 years	63.0	
				No screening		63.0	
ERSPC Finland	1996-	Finland	Men aged 55 to 67 without a previous diagnosis of prostate cancer	Screening	12 years	58.7	
				No screening		58.7	
ERSPC France	2000- (Tarn); 2003- (Herault)	France	Men aged 50 to 74 years	Screening	7.5 and 6.4 years	61.7 (Tarn) 62.6 (Herault)	
				No screening		61.7 (Tarn) 62.6 (Herault)	
ERSPC Italy	1996-	Italy	Men aged 55 to 70 years	Screening	12.6 years	61.8	
				No screening		61.8	

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Table 1. Overview of studies included in synthesis (study characteristics, intervention, and comparisons) *(Continued)*

ERSPC (Rotterdam) Netherlands	1993-	Netherlands	Men aged 54 to 74 years	Screening	13 years	63.2	
				No screening		63.2	
ERSPC Spain	1996-	Spain	Men aged 45 to 70 years with a life expectancy greater than 10 years	Screening	21.1 years	56.9	
				No screening		57.1	
ERSPC (Göteborg) Sweden	1994-	Sweden	Men aged 50 to 65 years without a current diagnosis of prostate cancer	Screening	18 years	56.4	
				No screening		56.4	
ERSPC (Aargau) Switzerland	1998-	Switzerland	Men aged 55 to 70 years	Screening	12.2 years	60.5	
				No screening		60.7	
Stockholm	1988-2008	Sweden	Men aged 55 to 70 years without a previous history of prostate cancer	Screening	20 years	62.4	PCa mortality, overall mortality, PCa diagnosis
				No screening		62.5	
Quebec	1988-2004	Canada	Men aged 45 to 80 years	Screening	11 years	NR	PCa mortality
				No screening		NR	
PLCO	1993-	USA	Men aged 55 to 74 years	Screening	15 years	NR (Reported as age category)	PCa mortality, overall mortality, PCa diagnosis, localised and advanced PCa diagnosis, metastatic PCa diagnosis
				No screening		NR (Reported as age category)	
ProScreen	2018-	Finland	Men aged 50 to 63 years, without a previous diagnosis of prostate cancer	Screening[a]	3.2 years[b]	57.3	PCa diagnosis in general, localised and advanced
				No screening		57.3	

[a]Screening using a stepwise approach with a kallikrein panel and magnetic resonance imaging for those with elevated prostate-specific antigen (PSA) test
[b]Expected follow-up: 15 years
PCa: prostate cancer

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Table 2. Participants in included studies

	Intervention(s) and comparator(s)	Eligible (N)	Randomised (N)	ITT (N)	Analysed (N)	Finishing trial (N)	Randomised finishing trial (%)
CAP	Screening	419582	195912	195912	189386	189386	96.6
	No screening	415357	219445	219445	219439	219439	99.9
	Total:		415357	415357	408825	408825	98.4
ERSPC[a]	Screening	182160	72888	72888	72888	72888	100
	No screening	162236	89348	89348	89348	89348	100
	Total:		162236	162236	162236	162236	100
ERSPC (Antwerp) Belgium	Screening	NR	4307	4307	4307	4307	100
	No screening	NR	4255	4255	4255	4255	100
	Total:		8562	8562	8562	8562	100
ERSPC Finland	Screening	NR	31866	31866	31866	31866	100
	No screening	80456	48278	48278	48278	48278	100
	Total:		80144	80144	80144	80144	100
ERSPC France	Screening	NR	42447	42447	42447	42447	100
	No screening	84781	42065	42065	42065	42065	100
	Total:		84512	84512	84512	84512	100
ERSPC Italy	Screening	NR	7266	7266	7266	7266	100
	No screening	NR	7251	7251	7251	7251	100
	Total:		14517	14517	14517	14517	100
ERSPC (Rotterdam) Netherlands	Screening	NR	21210	21210	21210	21210	100

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Table 2. Participants in included studies *(Continued)*

	No screening	88283	21166	21166	21166	21166	100
	Total:		42376	42376	42376	42376	100
ERSPC Spain	Screening	NR	2414	2414	2414	2414	100
	No screening	18612	1861	1861	1861	1861	100
	Total:		4275	4275	4275	4275	100
ERSPC (Göteborg) Sweden	Screening	NR	10000	9950	9950	9950	99.5
	No screening	32298	10000	9949	9949	9949	99.5
	Total:		20000	19899	19899	19899	99.5
ERSPC (Aargau) Switzerland	Screening	NR	5130	5130	5130	5130	100
	No screening	18361	5121	5121	5121	5121	100
	Total:		10251	10251	10251	10251	100
Stockholm	Screening	NR	2400	2400	2400	2400	100
	No screening	27481	25081	25081	25081	25081	100
	Total:		27146	27146	27146	27146	100
Quebec	Screening	NR	31133	31133	31133	31133	100
	No screening	46486	15353	15353	15353	15353	100
	Total:		46486	46486	46486	46486	100
PLCO	Screening	NR/	38340	38340	38340	38340	100
	No screening	76683	38343	38343	38343	38343	100
	Total:		76683	76683	76683	76683	100
ProScreen	Screening	60745	15299	15201	15201	15201	100[b]

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Table 2. Participants in included studies *(Continued)*

	No screening	45544	45544	45544	45544	100[b]
Total:		60843	60745	60745	60745	100
Grand totals	Screening	355972			349348	
	No screening	433114			433108	
	Overall	789086			782456	

ITT: intention-to-treat; **NR:** not reported
Only numbers from the 'parent' ERPSC study are included in the grand totals (i.e. not individual ERSPC sites).
[a]Numbers reflect the "core group"
[b]Preliminary results after the first round of screening

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